Planning and Reinforcement Learning in AI Systems

(PARLAI)

Lesson 4b - Model-based Planning - Fully Observable Stochastic

Sarah Keren

The Taub Faculty of Computer Science
Technion - Israel Institute of Technology

What is Stochastic Planning?

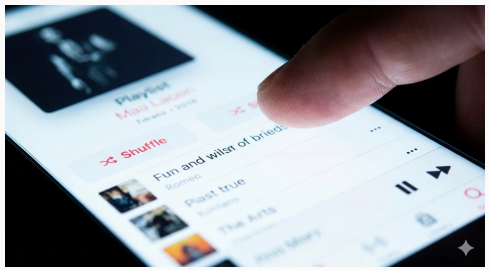
Stochastic Planning

Stochastic planning extends classical planning to settings where action outcomes are not fully predictable.

- The same action can lead to different outcomes
- We must reason about **uncertainty** and **risk**
- Solutions are **policies**, not fixed plans







What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

Do we need to model stochastic action effects?

If we use a deterministic model, we can compute a plan and replan if needed.

But replanning alone is not always enough or even possible:

- Outcomes may be **inherently uncertain**, so the same action can lead to different next states.
- Some actions may have **riskier** outcomes than others, even if they look equally good on average
- An agent must reason not only about the **next step**, but about the **distribution over future outcomes**

Deterministic planning finds a path.
Stochastic planning finds a policy that is robust to uncertainty.

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

Do we need to model stochastic action effects?

If we use a deterministic model, we can compute a plan and replan if needed.

But replanning alone is not always enough or even possible:

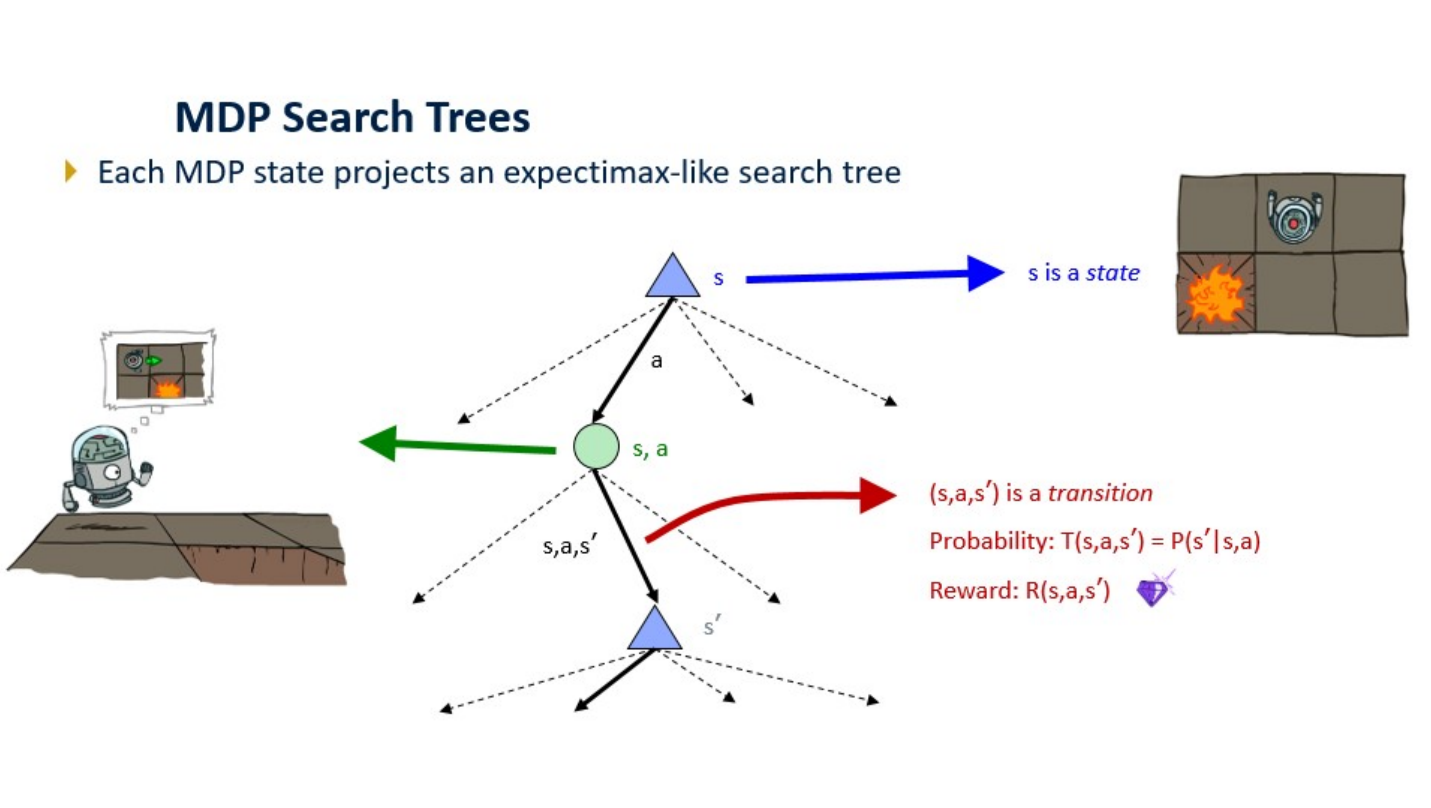






What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

Stochastic Planning from IAI



From IAI

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

Reminder: Stochastic fully observable environments

Stochastic fully observable environments are defined using a **Markov Decision Process** (MDP) $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$ where

- $\mathcal{S}$ is a finite set of states
- $\mathcal{A}$ is a finite set of actions
- $\mathcal{P}$ is a state transition probability matrix $\mathcal{P}^a_{s,s'} = \mathcal{P}[S_{t+1} = s' | S_t = s, A_t = a]$
- $\mathcal{R}$ is a reward function, $\mathcal{R}^a_s = \mathbb{E}[R_{t+1} | S_t = s, A_t = a]$, and
- **optional:** γ is a discount factor $\gamma \in [0, 1]$ that is used to favor immediate rewards over future rewards.

The Markov property: "The future is independent of the past given the present".

Extensions: Infinite and continuous MDPs, partially observable MDPs, undiscounted, average reward MDPs. etc.

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

Reminder: Markov Decision Process

A **Markov Decision Process**(MDP) is a tuple $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$ where

- $\mathcal{S}$ is a finite set of states, defined over a set of variables $\mathcal{X}$
- $\mathcal{A}$ is a finite set of actions
- $\mathcal{P}$ is a state transition probability matrix $\mathcal{P}^a_{s,s'} = \mathcal{P}[S_{t+1} = s' | S_t = s, A_t = a]$
- $\mathcal{R}$ is a reward function, $\mathcal{R}^a_s = \mathbb{E}[R_{t+1} | S_t = s, A_t = a]$, and
- **optional:** γ is a discount factor $\gamma \in [0, 1]$ that is used to favor immediate rewards over future rewards.

The Markov property: "The future is independent of the past given the present".

Extensions: Infinite and continuous MDPs, partially observable MDPs, undiscounted, average reward MDPs. etc.

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

Reminder: Value Functions

Since the rewards the agent can expect to receive in the future depend on what actions it will take, value functions are defined with respect to particular policies.

State-Value Function for Policy π:

$$V_\pi(s) \doteq \mathbb{E}_\pi \left[G_t \mid S_t = s \right]$$

Action-Value Function for Policy π:

$$Q_\pi(s, a) \doteq \mathbb{E}_\pi \left[G_t \mid S_t = s, A_t = a \right]$$

where G_t is the return, the total accumulated (typically discounted) reward obtained along a trajectory from time t onward.

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

Reminder: True vs. Estimated Values - Notations

True (Unknown) value functions are denoted as:

- $V_\pi(s)$ - true state value
- $Q_\pi(s, a)$ - true action value
- Defined via expectation:

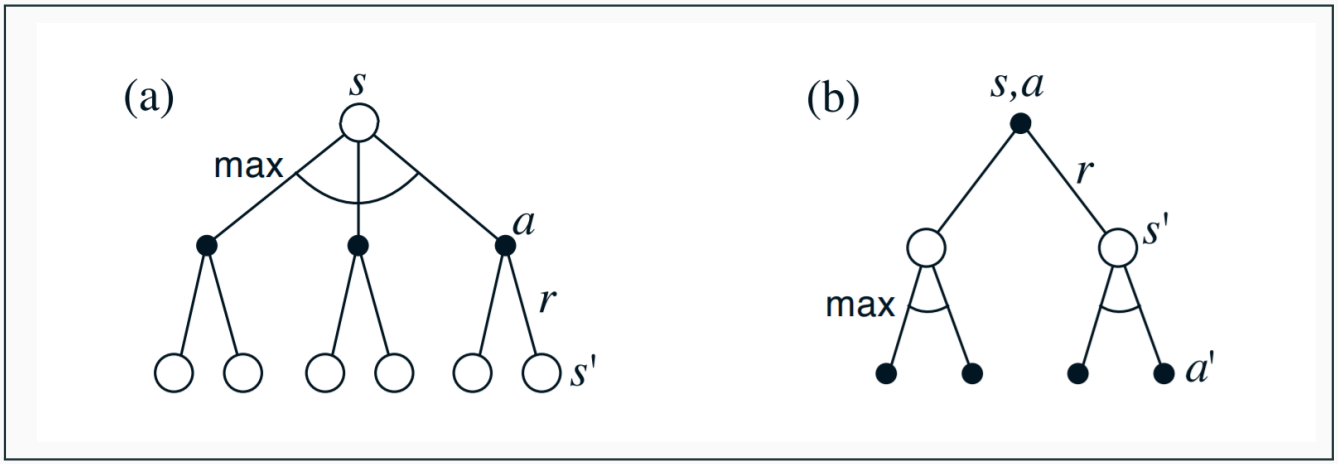
$$V_\pi(s) = \mathbb{E}_\pi[G_t \mid S_t = s]$$

Estimated (learned) value functions (updated over time) are denoted as:

- $\hat{V}(s)$ - estimated state value
- $\hat{Q}(s, a)$ - estimated action value

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

Reminder: Value Functions



Backup diagrams for $V_\pi(s)$ (left) and $Q_\pi(s, a)$ (right).

Since it may not be practical to keep separate averages for each state individually V_π and Q_π are often represented as parameterized functions (with fewer parameters than states).

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

And-Or Graphs

Deterministic planning induces a **search tree**, where each node represents an action choice of the agent. These selections are **Or** nodes in the tree.

For conditional / probabilistic planning we need to choose which action to take at every state (**Or**), but must handle every outcome for the action (**And**).

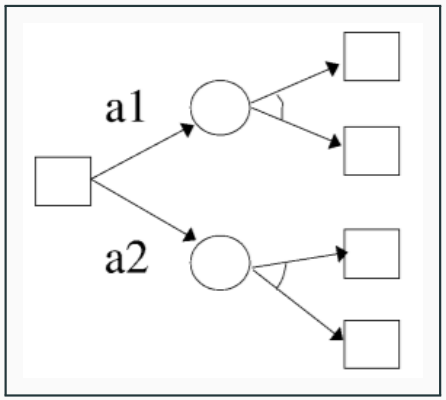
A solution is a **conditional plan / policy** rather than just a single move.

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

And-Or Graphs

For this purpose we use and-or-graphs with two kinds of nodes:

- **OR node** - specify a selection between actions
- **AND node** - specify the possible outcomes



Where have you already used an and-or graph?

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

AND-OR Graphs: Formal Definition

An **AND–OR graph** is a directed hypergraph

$$G = (V_{\text{OR}}, V_{\text{AND}}, E)$$

where:

- OR-nodes: represent **choices**.
- AND-nodes: represent **subtask decompositions**.
- Hyperedge $(v, C) \in E$: node v expands to children C.

Goal: Find a **solution graph**, i.e., a subgraph representing a minimal cost solution reachable from s_0 containing:

- one successor for each OR node,
- all successors for each AND node

What do the hyperedges represent?

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

AND-OR Graphs: Formal Definition

- For an OR node s with successors s':

$$V(s) = \max_{s' \in \mathrm{Succ}(s)} \left[\mathcal{R}(s, s') + V(s') \right]$$

  (choose the best option)

- For an AND node s with successors $s_1, \ldots, s_k$:

$$V(s) = \sum_{i=1}^{k} \left[\mathcal{R}(s, s_i) + V(s_i) \right]$$

  (all subproblems must be solved- here, result is their summation)

Other optional operators for OR and AND ?

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

And-Or Graphs Example: Deterministic vacuum world

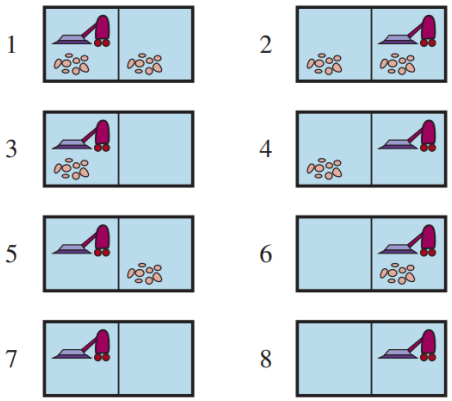


Figure 4.8 The eight possible states of the vacuum world; states 7 and 8 are goal states.

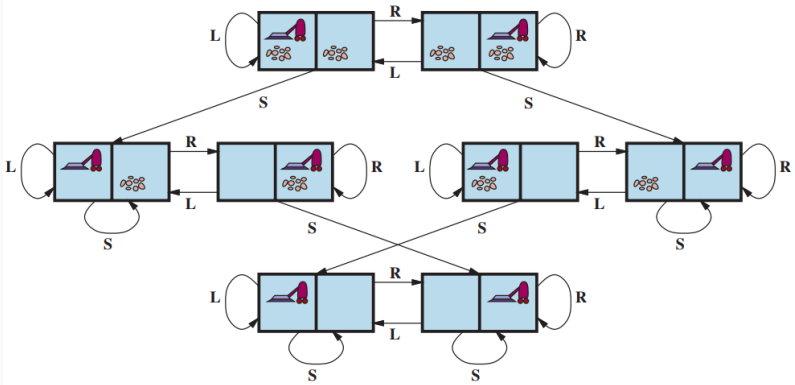


Figure 3.2 The state-space graph for the two-cell vacuum world. There are 8 states and three actions for each state: L = *Left*, R = *Right*, S = *Suck*.

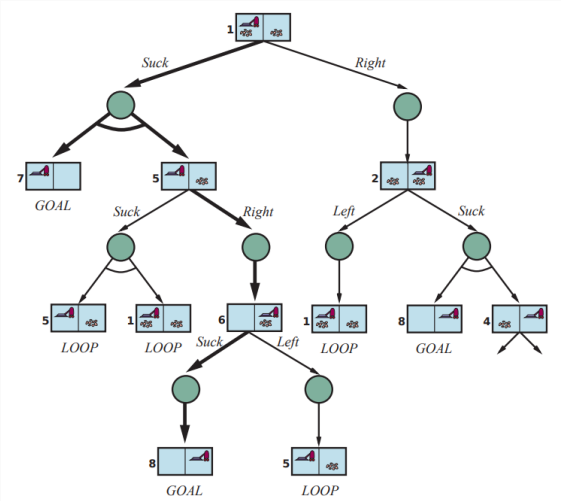
From Artificial Intelligence A Modern Approach by Russel and Norvig

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

And-Or Graphs Example: Erratic vacuum world

In this version, when Suck is applied to a dirty square, the resulting action cleans the square and sometimes also cleans an adjacent square. When applied to a clean square, this sometimes deposits dirt on the carpet.

How to compute a policy?



From Artificial Intelligence A Modern Approach by Russel and Norvig

What is Stochastic Planning?

Challenges

Solution Approaches

The Bellman Equation

Dynamic Programming Methods

Heuristic Search in Stochastic Domains

Conclusion

Challenges

Challenges in Stochastic Planning

In addition to the challenges encountered in deterministic planning, stochastic planning introduces several fundamental challenges:

- Outcome uncertainty
- Policy (instead of plan) computation
- Search-space explosion

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

Challenges in Stochastic Planning

Outcomes are not just multiple - they are **probabilistic**
$\rightarrow$ Need to reason about **expected return** and must balance **risk vs reward**.

Each action leads to **multiple possible outcomes**. The search tree becomes an **AND-OR graph** with exponential growth.

We need more efficient frameworks for reasoning under uncertainty.

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

Per-Step Complexity in Stochastic Planning

For each state $s \in \mathcal{S}$, we consider every action $a \in$.

Deterministic case:

- We evaluate one successor per action
- Time complexity: $\mathcal{O}(|\mathcal{A}|)$

Stochastic case:

- Each action a may lead to multiple successor states $\rightarrow$ we must consider all $s' \in \mathcal{S}$ such that $\mathcal{P}(s, a, s') > 0$
- If we compute expectation over next states:

$$\sum_{s'} \mathcal{P}(s, a, s') V(s')$$

- Time complexity: $\mathcal{O}(|\mathcal{A}| \cdot |\mathcal{S}|)$

Each decision requires reasoning over **all possible next states**.

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

Overall Complexity in Stochastic Planning

- **Per-state computation:**
  - evaluating all actions and outcomes:
    $$\mathcal{O}(|\mathcal{A}| \cdot |\mathcal{S}|)$$
- **Across all states:**
  - must compute values / decisions for each state.
  - total cost:
    $$\mathcal{O}(|\mathcal{S}| \cdot |\mathcal{A}| \cdot |\mathcal{S}|) = \mathcal{O}(|\mathcal{S}|^2 \cdot |\mathcal{A}|)$$
- **Planning horizon / iterations:**
  - may require multiple passes: repeat updates until convergence or horizon K.
  - total complexity:
    $$\mathcal{O}(K \cdot |\mathcal{S}|^2 \cdot |\mathcal{A}|)$$
    where K is the number of iterations.
  - K varies between approaches and depends on discount factor / accuracy / horizon.

Polynomial in $|\mathcal{S}|$ does not imply tractability: $|\mathcal{S}|$ is exponential

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

Solution Approaches

Solution Approaches

- Optimistic planning with replanning
- Non-deterministic planning (ignoring probabilities)
- Stochastic planning

The different approaches trade off **planning effort** and **solution quality**.

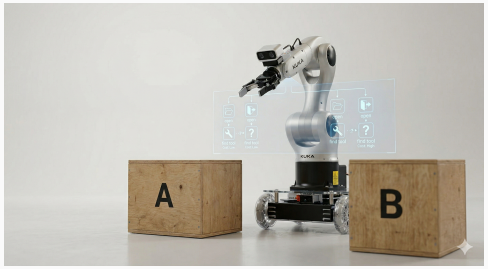
What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

Replanning

- Solve a simplified or deterministic relaxation of the problem.
    - A common approach: plan under a single hypothesized state – often MAP (Maximum A Posteriori)
- Execute the resulting plan until unexpected outcomes occur.
- Replan from the new state when needed.

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

Replanning: Example



- Assume a single hypothesis (e.g., tool is in Box A).
- Plan: open Box A $\rightarrow$ pick-up tool.
- If observation contradicts assumption, replan.

Simple and scalable, but reactive (rather than deliberative) to uncertainty.

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

Non-Deterministic Planning

- Construct a **conditional plan**
- Plan for all/some possible outcomes of an action
  - actions may lead to multiple possible successor states
  - the agent branches its behavior based on observations
- Solutions are often represented as **AND-OR trees**
  - OR nodes: choose an action
  - AND nodes: handle all possible outcomes

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

Non-Deterministic AND-OR Search

AND-OR-SEARCH

OR-SEARCH($problem, state, path$) ▷ returns a conditional plan or failure
  if IS-GOAL($state$) then return empty plan
  if IS-CYCLE($path$) then return failure
  for each $action \in$ ACTIONS($state$):
    $plan \leftarrow$ AND-SEARCH($problem$, RESULTS($state, action$), $[state] + path$)
    if $plan \neq failure$ then return $[action] + plan$
  return failure

AND-SEARCH($problem, states, path$) ▷ returns a conditional plan or failure
  for each $s_i \in states$:
    $plan_i \leftarrow$ OR-SEARCH($problem, s_i, path$)
    if $plan_i = failure$ then return failure
  return [if s_1 then $plan_1$ else $\cdots$ else $plan_n$]

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

Non-Deterministic AND-OR Search

AND-OR-SEARCH

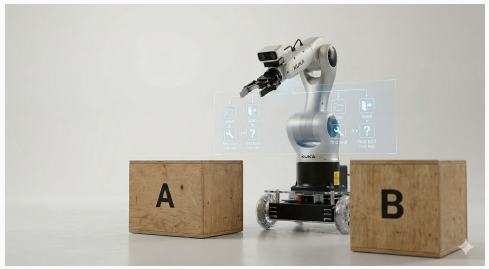
OR-SEARCH($problem, state, path$) ▷ returns a conditional plan or failure
  if IS-GOAL($state$) then return empty plan
  if IS-CYCLE($path$) then return failure
  for each $action \in$ ACTIONS($state$):
    $plan \leftarrow$ AND-SEARCH($problem$, RESULTS($state, action$), $[state] + path$)
    if $plan \neq failure$ then return $[action] + plan$
  return failure

AND-SEARCH($problem, states, path$) ▷ returns a conditional plan or failure
  for each $s_i \in states$:
    $plan_i \leftarrow$ OR-SEARCH($problem, s_i, path$)
    if $plan_i = failure$ then return failure
  return [if s_1 then $plan_1$ else $\cdots$ else $plan_n$]

What if we have probabilistic effects?

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

Non-Deterministic Planning: Example



- Construct a **conditional plan**
- Plan:
  - open Box A
  - **if** tool found → pick-up
  - **else** → open Box B → pick-up

Contingent & complete

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

Systematic Approaches to Planning with MDPs

Dynamic Programming – example algorithms:

- value iteration
- policy iteration

Heuristic Search – example algorithms:

- AO*, LAO*
- RTDP

Hybrid Approaches (Planning + Replanning):

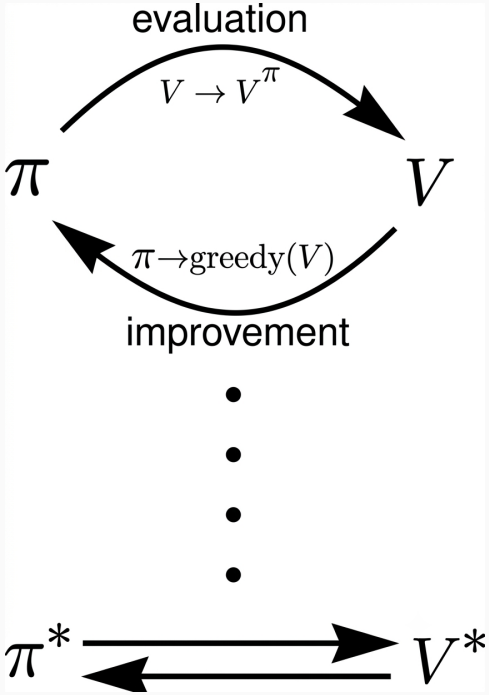
- Combine offline deliberation with online adaptation.

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

Reminder: Evaluation and Control

Prediction / Evaluation: evaluate the future given a policy - evaluate the quality of a given policy

Control/ Update / Improvement: optimize the policy.



What is Stochastic Planning?

Challenges

Solution Approaches

The Bellman Equation

Dynamic Programming Methods

Heuristic Search in Stochastic Domains

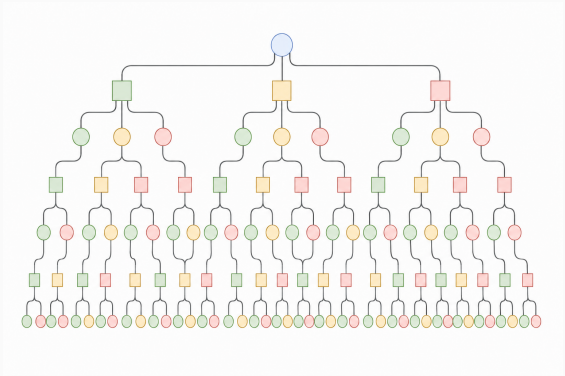
Conclusion

An Exhaustive Approach

For a policy π $V_\pi(s) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t R_t \mid S_0 = s\right]$

Naive computation: Compute the value of a state by **enumerating possible futures**.

- enumerate all possible trajectories starting from s
- compute total discounted reward for each trajectory
- take expectation over all trajectories

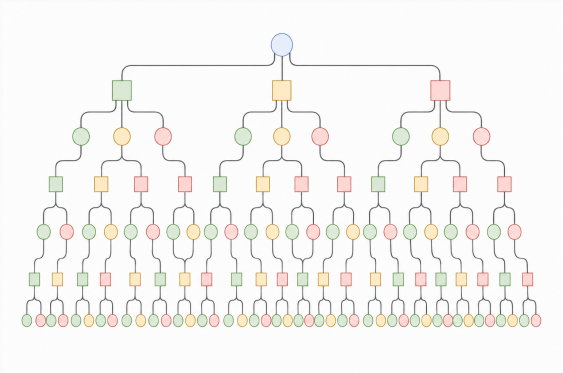


This requires reasoning over an exponentially large tree of futures.

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

An Exhaustive Approach

- The number of trajectories grows exponentially with the horizon
- Repeated subproblems (same states appear many times)



Can we avoid recomputing the value of the same state?

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

From Enumeration to Bootstrapping

- Instead of expanding full trajectories:
  - reuse value estimates of successor states
- **Bootstrapping:**
  - estimate value using **other value estimates**
  - $V(s)$ depends on $V(s')$ instead of full rollouts

Replace exponential trajectory enumeration with recursive value decomposition.

This leads to the Bellman equation.

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

The Bellman Equation

The Bellman Equation (from Sutton and Barto 2018)

A fundamental property of value functions used throughout dynamic programming and reinforcement learning is that they satisfy particular recursive relationships.

$$\text{value} = \text{reward} + \text{discount factor} \times \text{next value}$$

$$V_\pi(s) = \sum_{s'} \mathcal{P}(s' \mid s, \pi(s)) \left[\mathcal{R}(s, \pi(s), s') + \gamma V^\pi(s')\right]$$

If the reward function and policy are stochastic, then

$$V_\pi(s) = \sum_a \pi(a|s) \sum_{s'} \sum_r p(s', r|s, a) \left[r + \gamma V_\pi(s')\right], \forall s \in \mathcal{S}$$

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

The Bellman Equation (from Sutton and Barto 2018)

For any policy π and any state s, the following consistency conditions hold between the value of s and the value of its possible successors:

$$V_\pi(s) \doteq \mathbb{E}_\pi \left[G_t \mid S_t = s \right]$$

What is Stochastic Planning?

Challenges

Solution Approaches

The Bellman Equation

Dynamic Programming Methods

Heuristic Search in Stochastic Domains

Conclusion

The Bellman Equation (from Sutton and Barto 2018)

For any policy π and any state s, the following consistency conditions hold between the value of s and the value of its possible successors:

$$V_\pi(s) \doteq \mathbb{E}_\pi \left[G_t \mid S_t = s \right]$$

$$= \mathbb{E}_\pi \left[\sum_{k=0}^{\infty} \gamma^k r_{t+k+1} \mid S_t = s \right]$$

What is Stochastic Planning?

Challenges

Solution Approaches

The Bellman Equation

Dynamic Programming Methods

Heuristic Search in Stochastic Domains

Conclusion

The Bellman Equation (from Sutton and Barto 2018)

For any policy π and any state s, the following consistency conditions hold between the value of s and the value of its possible successors:

$$V_\pi(s) \doteq \mathbb{E}_\pi \left[G_t \mid S_t = s \right]$$

$$= \mathbb{E}_\pi \left[\sum_{k=0}^{\infty} \gamma^k r_{t+k+1} \mid S_t = s \right]$$

$$= \mathbb{E}_\pi \left[r_{t+1} + \gamma G_{t+1} \mid S_t = s \right]$$

What is Stochastic Planning?

Challenges

Solution Approaches

The Bellman Equation

Dynamic Programming Methods

Heuristic Search in Stochastic Domains

Conclusion

The Bellman Equation (from Sutton and Barto 2018)

For any policy π and any state s, the following consistency conditions hold between the value of s and the value of its possible successors:

$$V_\pi(s) \doteq \mathbb{E}_\pi \left[G_t \mid S_t = s \right]$$

$$= \mathbb{E}_\pi \left[\sum_{k=0}^{\infty} \gamma^k r_{t+k+1} \mid S_t = s \right]$$

$$= \mathbb{E}_\pi \left[r_{t+1} + \gamma G_{t+1} \mid S_t = s \right]$$

$$= \mathbb{E}_\pi \left[r_{t+1} + \gamma \sum_{k=0}^{\infty} \gamma^k r_{t+k+2} \mid S_t = s \right]$$

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

The Bellman Equation (from Sutton and Barto 2018)

For any policy π and any state s, the following consistency conditions hold between the value of s and the value of its possible successors:

$$V_\pi(s) \doteq \mathbb{E}_\pi \left[G_t \mid S_t = s \right]$$

$$= \mathbb{E}_\pi \left[\sum_{k=0}^{\infty} \gamma^k r_{t+k+1} \mid S_t = s \right]$$

$$= \mathbb{E}_\pi \left[r_{t+1} + \gamma G_{t+1} \mid S_t = s \right]$$

$$= \mathbb{E}_\pi \left[r_{t+1} + \gamma \sum_{k=0}^{\infty} \gamma^k r_{t+k+2} \mid S_t = s \right]$$

$$= \sum_a \pi(a|s) \sum_{s'} \sum_r p(s', r|s, a) \left[r + \gamma \mathbb{E}_\pi \left[\sum_{k=0}^{\infty} \gamma^k R_{t+k+2} \mid S_{t+1} = s' \right] \right]$$

What is Stochastic Planning?

Challenges

Solution Approaches

The Bellman Equation

Dynamic Programming Methods

Heuristic Search in Stochastic Domains

Conclusion

The Bellman Equation (from Sutton and Barto 2018)

For any policy π and any state s, the following consistency conditions hold between the value of s and the value of its possible successors:

$$V_\pi(s) \doteq \mathbb{E}_\pi \left[G_t \mid S_t = s \right]$$

$$= \mathbb{E}_\pi \left[\sum_{k=0}^{\infty} \gamma^k r_{t+k+1} \mid S_t = s \right]$$

$$= \mathbb{E}_\pi \left[r_{t+1} + \gamma G_{t+1} \mid S_t = s \right]$$

$$= \mathbb{E}_\pi \left[r_{t+1} + \gamma \sum_{k=0}^{\infty} \gamma^k r_{t+k+2} \mid S_t = s \right]$$

$$= \sum_a \pi(a|s) \sum_{s'} \sum_r p(s', r|s, a) \left[r + \gamma \mathbb{E}_\pi \left[\sum_{k=0}^{\infty} \gamma^k R_{t+k+2} \mid S_{t+1} = s' \right] \right]$$

$$= \sum_a \pi(a|s) \sum_{s'} \sum_r p(s', r|s, a) \left[r + \gamma \mathbb{E}_\pi \left[G_{t+1} \mid S_{t+1} = s' \right] \right]$$

What is Stochastic Planning?

Challenges

Solution Approaches

The Bellman Equation

Dynamic Programming Methods

Heuristic Search in Stochastic Domains

Conclusion

The Bellman Equation (from Sutton and Barto 2018)

For any policy π and any state s, the following consistency conditions hold between the value of s and the value of its possible successors:

$$V_\pi(s) \doteq \mathbb{E}_\pi \left[G_t \mid S_t = s \right]$$

$$= \mathbb{E}_\pi \left[\sum_{k=0}^{\infty} \gamma^k r_{t+k+1} \mid S_t = s \right]$$

$$= \mathbb{E}_\pi \left[r_{t+1} + \gamma G_{t+1} \mid S_t = s \right]$$

$$= \mathbb{E}_\pi \left[r_{t+1} + \gamma \sum_{k=0}^{\infty} \gamma^k r_{t+k+2} \mid S_t = s \right]$$

$$= \sum_a \pi(a|s) \sum_{s'} \sum_r p(s', r|s, a) \left[r + \gamma \mathbb{E}_\pi \left[\sum_{k=0}^{\infty} \gamma^k R_{t+k+2} \mid S_{t+1} = s' \right] \right]$$

$$= \sum_a \pi(a|s) \sum_{s'} \sum_r p(s', r|s, a) \left[r + \gamma \mathbb{E}_\pi \left[G_{t+1} \mid S_{t+1} = s' \right] \right]$$

$$= \sum_a \pi(a|s) \sum_{s'} \sum_r p(s', r|s, a) \left[r + \gamma v_\pi(s') \right], \forall s \in \mathcal{S}$$

The above formulations are the basis of different ways to compute V_π.

What is Stochastic Planning?

Challenges

Solution Approaches

The Bellman Equation

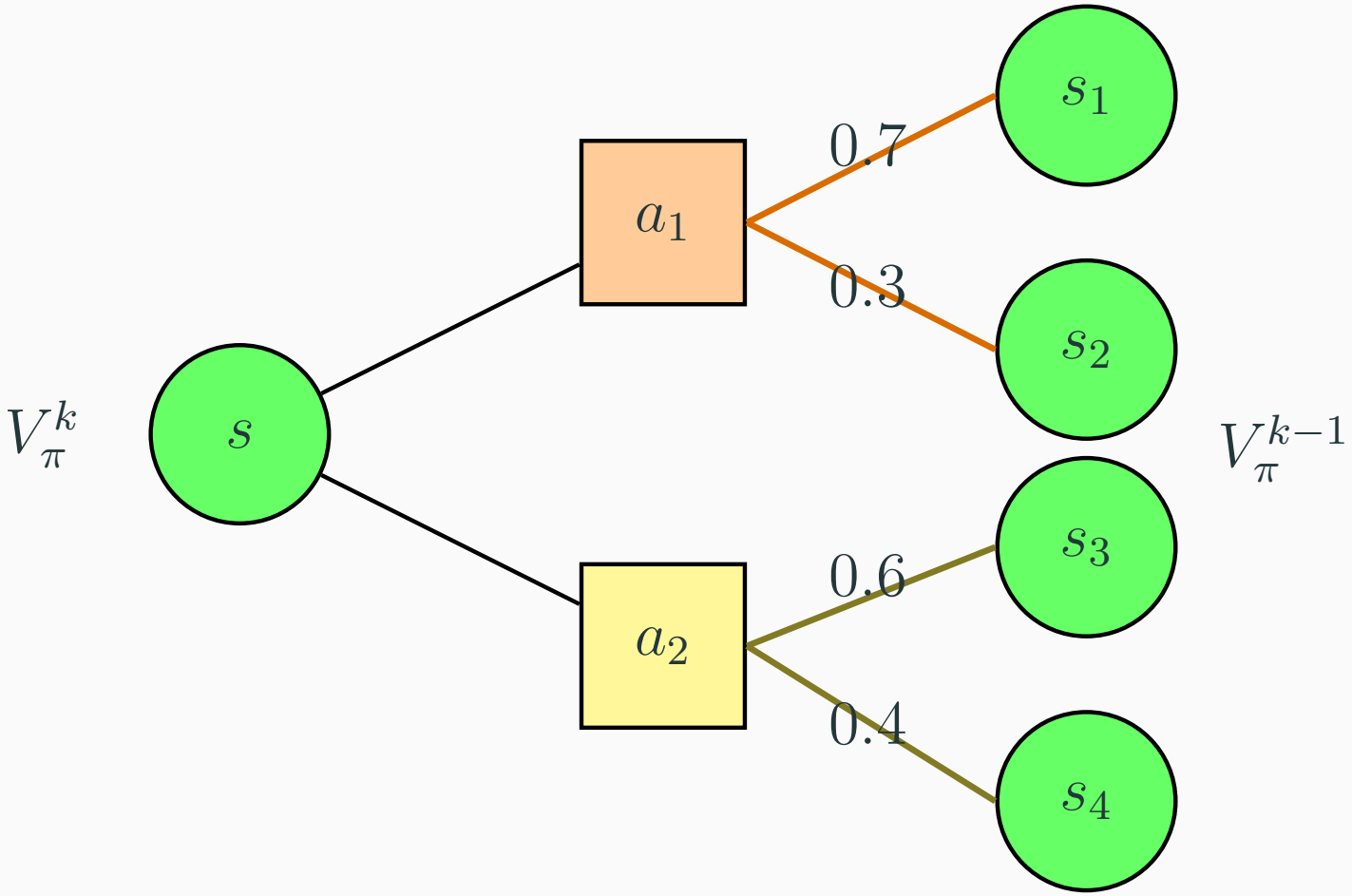
Dynamic Programming Methods

Heuristic Search in Stochastic Domains

Conclusion

Bellman Backups

How can we compute $V_\pi^k(s)$ given $V_\pi^{k-1}(s)$?



What is Stochastic Planning?

Challenges

Solution Approaches

The Bellman Equation

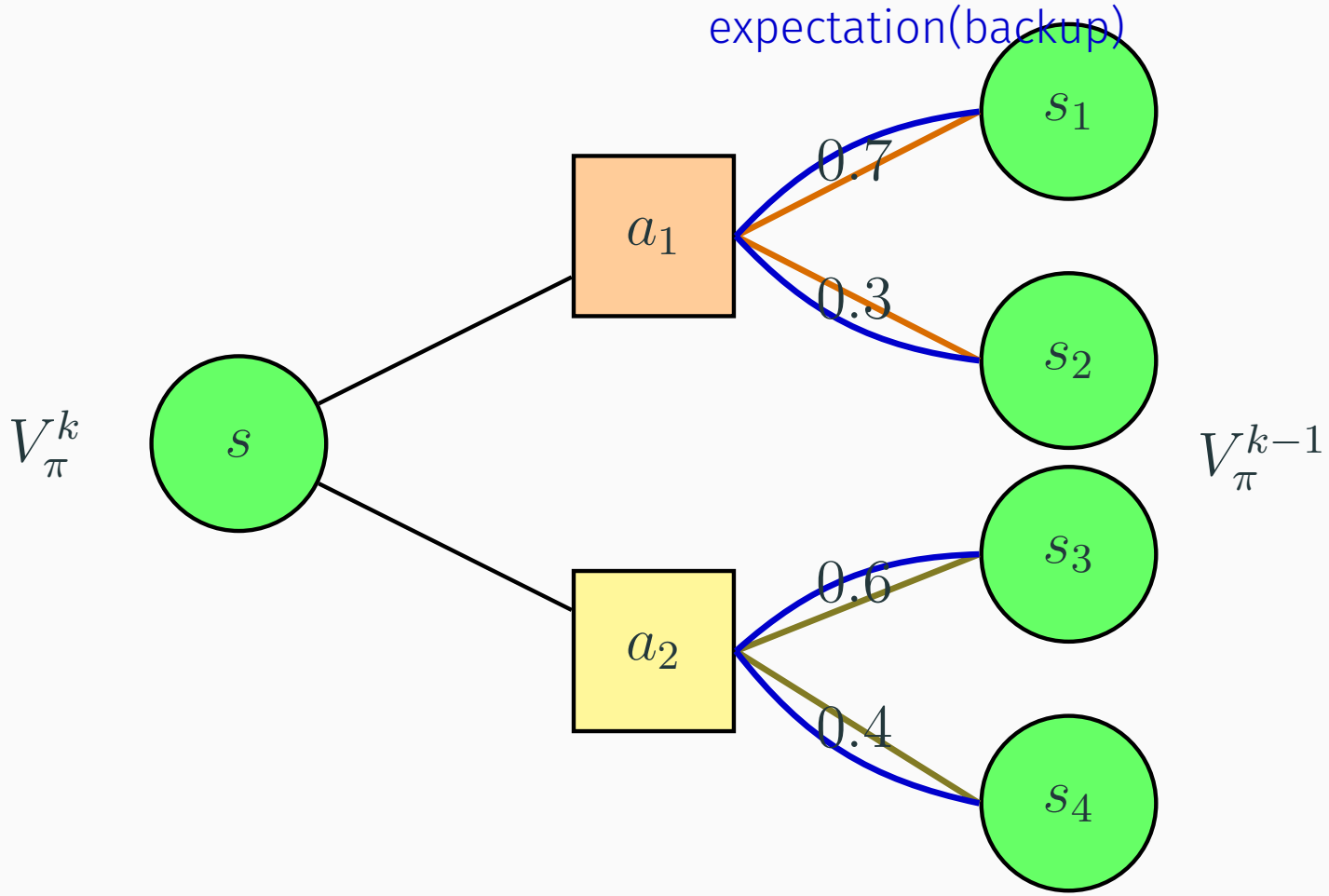
Dynamic Programming Methods

Heuristic Search in Stochastic Domains

Conclusion

Bellman Backups

How can we compute $V_\pi^k(s)$ given $V_\pi^{k-1}(s)$?



What is Stochastic Planning?

Challenges

Solution Approaches

The Bellman Equation

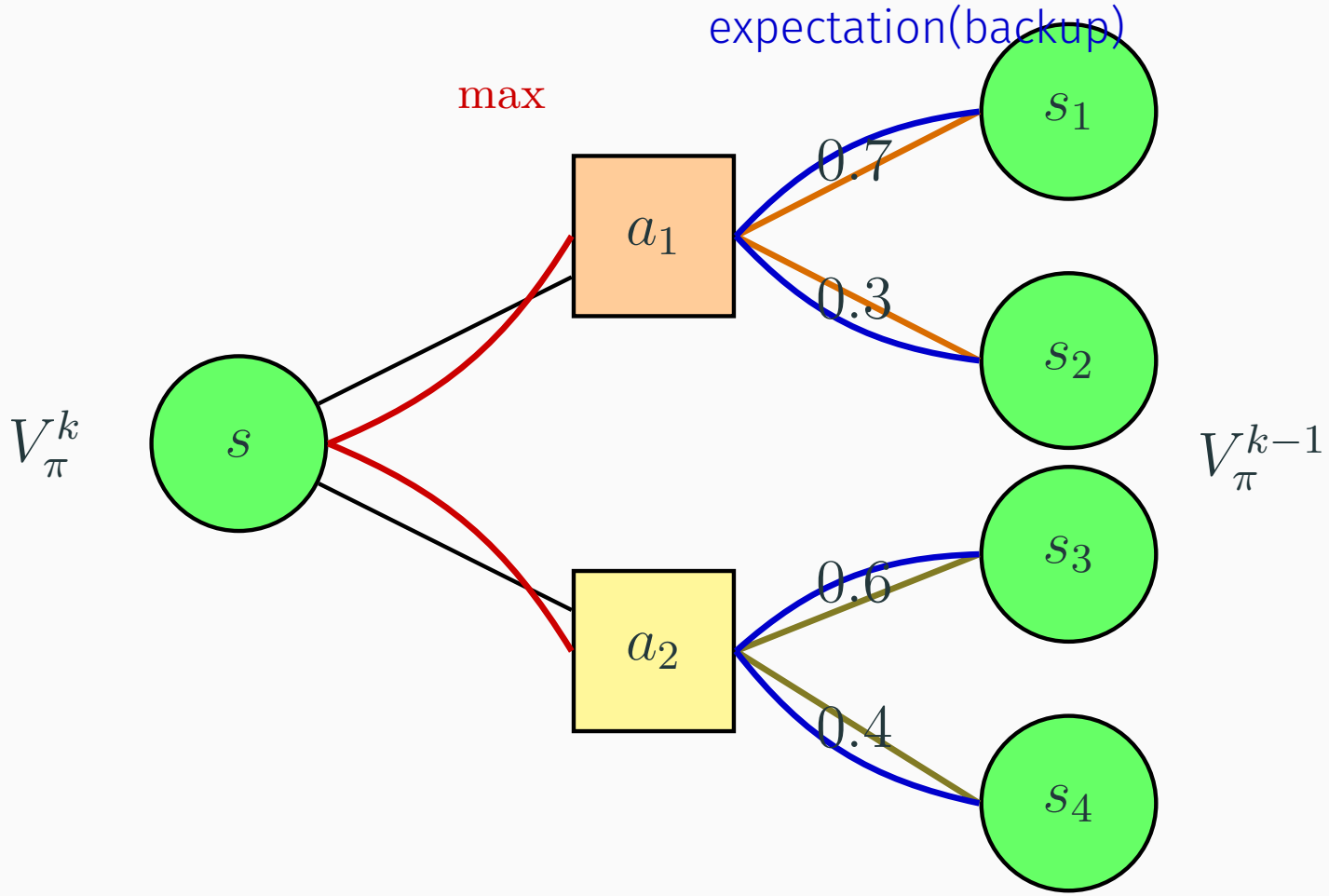
Dynamic Programming Methods

Heuristic Search in Stochastic Domains

Conclusion

Bellman Backups

How can we compute $V_\pi^k(s)$ given $V_\pi^{k-1}(s)$?



What is Stochastic Planning?

Challenges

Solution Approaches

The Bellman Equation

Dynamic Programming Methods

Heuristic Search in Stochastic Domains

Conclusion

The Bellman Principle of Optimality

An optimal solution can be constructed from optimal subsolutions.

- The value at a state is determined by **optimal choices** at that state
- Each choice leads to future states whose values must also be **optimal**
- This induces a **recursive structure** over the state space

Future values propagate backward (Bellman backup)

$\Downarrow$

Optimal decisions are made locally (max over actions)

This leads to the **Bellman optimality equation**

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

The Bellman Principle of Optimality

$$V^*(s) = \max_a \sum_{s'} P(s'|s,a) \left[R(s,a,s') + \gamma V^*(s') \right]$$

Equivalently: the value of a state equals the expected return from the **best action**, followed by optimal behavior.

$$V^*(s) = \max_{a \in A(s)} Q^*(s,a)$$

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

The Bellman Optimality Equation

$$V^*(s)$$

What is Stochastic Planning?

Challenges

Solution Approaches

The Bellman Equation

Dynamic Programming Methods

Heuristic Search in Stochastic Domains

Conclusion

The Bellman Optimality Equation

$$V^*(s)$$

max Q over actions:

$$= \max_{a \in A(s)} Q^*(s, a)$$

What is Stochastic Planning?

Challenges

Solution Approaches

The Bellman Equation

Dynamic Programming Methods

Heuristic Search in Stochastic Domains

Conclusion

The Bellman Optimality Equation

$$V^*(s)$$

max Q over actions:

$$= \max_{a \in A(s)} Q^*(s, a)$$

expected return from step t onward:

$$= \max_a \mathbb{E}_{\pi*} \left[G_t | S_t = s, A_t = a \right]$$

What is Stochastic Planning?

Challenges

Solution Approaches

The Bellman Equation

Dynamic Programming Methods

Heuristic Search in Stochastic Domains

Conclusion

The Bellman Optimality Equation

$$V^*(s)$$

max Q over actions:

$$= \max_{a \in A(s)} Q^*(s, a)$$

expected return from step t onward:

$$= \max_a \mathbb{E}_{\pi *} \left[G_t | S_t = s, A_t = a \right]$$

expected return from step t onward (k counts future steps):

$$= \max_a \mathbb{E}_{\pi *} \left[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1} | S_t = s, A_t = a \right]$$

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

The Bellman Optimality Equation

the maximal expected return for taking a at s at time t , receiving R_{t+1} and thereafter following an optimal:

$$= \max_a \mathbb{E}\left[R_{t+1} + \gamma V^*(S_{t+1}) | S_t = s, A_t = a\right]$$

What is Stochastic Planning?

Challenges

Solution Approaches

The Bellman Equation

Dynamic Programming Methods

Heuristic Search in Stochastic Domains

Conclusion

The Bellman Optimality Equation

the maximal expected return for taking a at s at time t , receiving R_{t+1} and thereafter following an optimal:

$$= \max_a \mathbb{E}\left[R_{t+1} + \gamma V^*(S_{t+1}) | S_t = s, A_t = a\right]$$

the maximal expectation over next state s' return r when taking a at s, computed as the immediate return r plus the discounted optimal value of s':

$$= \max_{a \in A(s)} \sum_{s',r} \mathcal{P}(s', r|s, a) \left[r + \gamma V^*(s')\right]$$

Bellman optimality (considering state transition uncertainty and reward uncertainty separately):

$$V^*(s) = \max_a \sum_{s'} \mathcal{P}(s'|s, a) \left[\mathcal{R}(s, a, s') + \gamma V^*(s')\right]$$

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

The Bellman Optimality Equation

the maximal expected return for taking a at s at time t , receiving R_{t+1} and thereafter following an optimal:

$$= \max_a \mathbb{E}\left[R_{t+1} + \gamma V^*(S_{t+1}) | S_t = s, A_t = a\right]$$

the maximal expectation over next state s' return r when taking a at s, computed as the immediate return r plus the discounted optimal value of s':

$$= \max_{a \in A(s)} \sum_{s',r} \mathcal{P}(s', r|s, a) \left[r + \gamma V^*(s')\right]$$

Bellman optimality (considering state transition uncertainty and reward uncertainty separately):

$$V^*(s) = \max_a \sum_{s'} \mathcal{P}(s'|s, a) \left[\mathcal{R}(s, a, s') + \gamma V^*(s')\right]$$

similar development for q functions:

$$Q^*(s, a) = \sum_{s',r} \mathcal{P}(s', r|s, a) \left[r + \gamma \max_{a'} Q^*(s', a')\right]$$

$$Q^*(s, a) = \sum_{s'} \mathcal{P}(s'|s, a) \left[\mathcal{R}(s, a, s') + \gamma \max_{a'} Q^*(s', a')\right]$$

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

The Bellman Optimality Equation

Why do we need so many versions of the same equation?

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

Dynamic Programming Methods

From Optimality Equations to Update rules

$$V^*(s) = \max_a \sum_{s'} \mathcal{P}(s'|s,a) \left[\mathcal{R}(s,a,s') + \gamma V^*(s')\right] \tag{1}$$

How can the Bellman optimality equation be used within a solution algorithm ?

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

From Optimality Equations to Update rules

$$V^*(s) = \max_a \sum_{s'} \mathcal{P}(s'|s,a) \left[\mathcal{R}(s,a,s') + \gamma V^*(s')\right] \tag{1}$$

How can the Bellman optimality equation be used within a solution algorithm ?

$$V_{k+1}(s) = \max_a \sum_{s'} \mathcal{P}(s'|s,a) \left[\mathcal{R}(s,a,s') + \gamma V_k(s')\right]$$

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

From Optimality Equations to Update rules

In many places in the literature, you might see this.

$$V(s) = \max_a (R(s,a) + \gamma V(s'))$$

What's missing here ?

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

From Optimality Equation to Policy Optimization

The Bellman optimality equation defines the value of an optimal policy:

$$V^*(s) = \max_a \sum_{s'} \mathcal{P}(s'|s,a) \left[\mathcal{R}(s,a,s') + \gamma V^*(s')\right]$$

But how do we compute V^*?

What is Stochastic Planning?

Challenges

Solution Approaches

The Bellman Equation

Dynamic Programming Methods

Heuristic Search in Stochastic Domains

Conclusion

From Optimality Equation to Policy Optimization

The Bellman optimality equation defines the value of an optimal policy:

$$V^*(s) = \max_a \sum_{s'} \mathcal{P}(s'|s,a) \left[\mathcal{R}(s,a,s') + \gamma V^*(s')\right]$$

But how do we compute V^*?

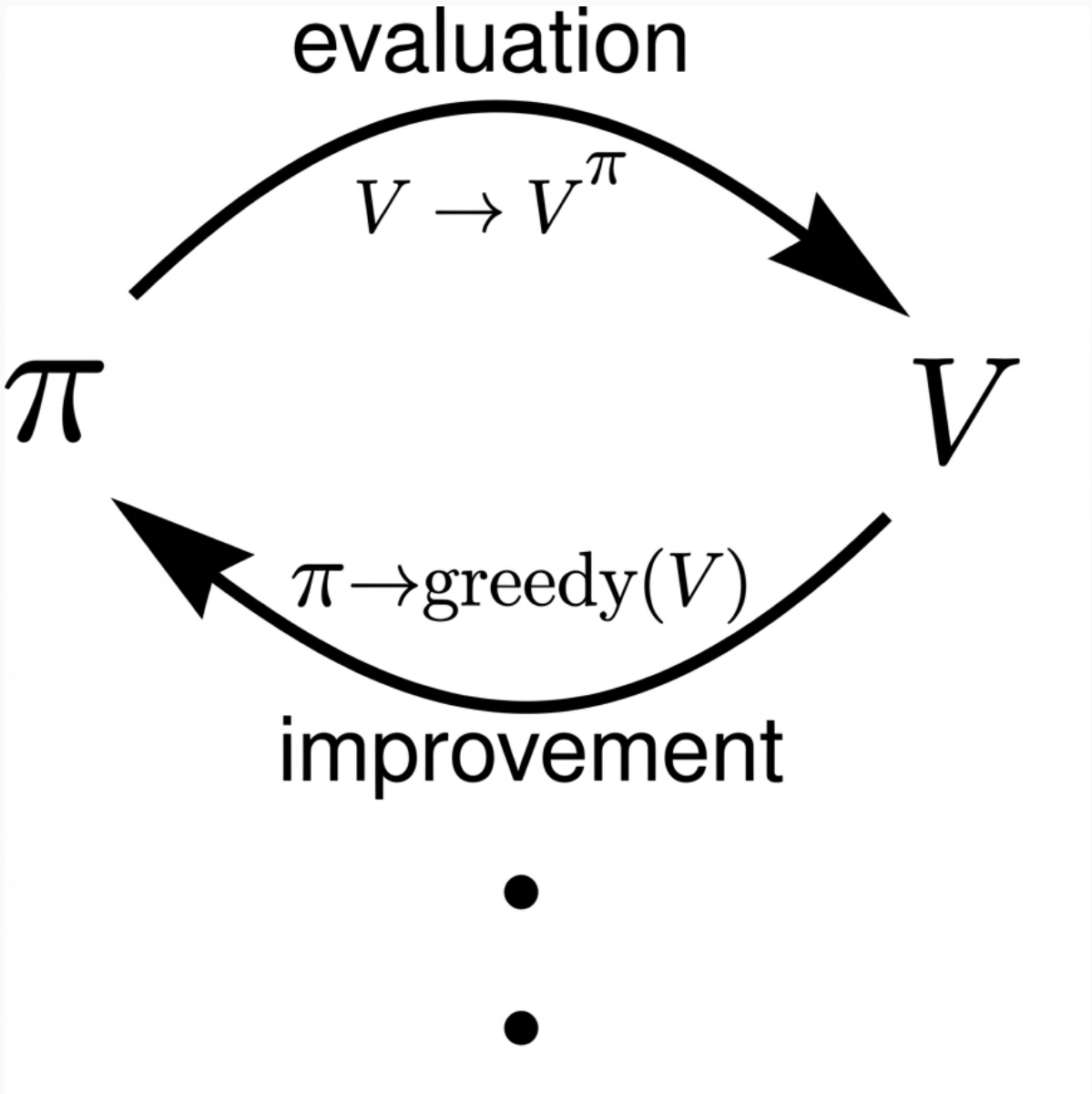
Key idea:

- Optimal solutions can be computed **recursively**
- This enables **dynamic programming algorithms**
- An iterative computation:
  - start with an initial guess V^0
  - repeatedly apply Bellman updates

This leads to algorithms that are based on Bellman backups,

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

The General Policy Iteration (GPI) Structure



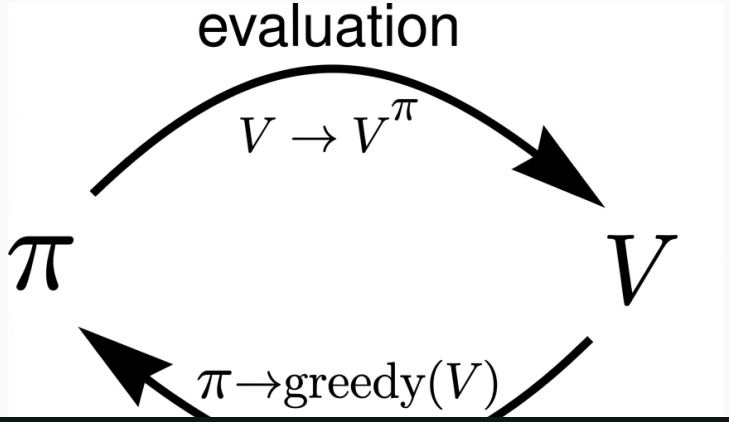
What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

Generalized Policy Iteration (GPI)

Most solution approaches can be characterized by two simultaneous, interacting processes:

1. **Policy evaluation -** making the value function consistent with the current policy.
2. **Policy improvement -** making the policy greedy with respect to the current value function.

Generalized Policy Iteration is the general idea of allowing policy evaluation and policy improvement to interact, independent of the granularity and other details of the processes.



What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

GPI: Policy Iteration and Value Iteration

Policy Iteration

- Given a policy π^k, we use V_π to improve the policy, and yield π^{k+1}. We then compute $V_{\pi^{k+1}}$ to yield an even better policy, π^{k+2}, etc.
- We can thus obtain a sequence of monotonically improving policies and value functions.

Value Iteration

- Until convergence, iterativly update values based on the value of the best next-state.
- Value iteration is obtained simply by turning the Bellman optimality equation into an update rule.

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

Policy Iteration

From IAI:

function POLICY-ITERATION(mdp) **returns** a policy
  inputs: mdp, an MDP with states S, actions $A(s)$, transition model $P(s' \mid s, a)$
  local variables: U, a vector of utilities for states in S, initially zero
    π, a policy vector indexed by state, initially random

  repeat
    $U \leftarrow$ POLICY-EVALUATION(π, U, mdp)
    $unchanged? \leftarrow$ true
    for each state s **in** S **do**
      $a^* \leftarrow \underset{a \in A(s)}{\text{argmax}}$ Q-VALUE(mdp, s, a, U)
      if Q-VALUE(mdp, s, a^*, U) $>$ Q-VALUE($mdp, s, \pi[s], U$) **then**
        $\pi[s] \leftarrow a^*$; $unchanged? \leftarrow$ false
  until $unchanged?$
  return π

What is Stochastic Planning?

Challenges

Solution Approaches

The Bellman Equation

Dynamic Programming Methods

Heuristic Search in Stochastic Domains

Conclusion

Policy Iteration

Policy Iteration (input: MDP)

Initialize arbitrary policy π_0

repeat

- **Policy Evaluation:**
  Compute $V^{\pi_k}(s)$ for all $s \in \mathcal{S}$:
  $$V^{\pi_k}(s) = R(s, \pi_k(s)) + \gamma \sum_{s'} P(s' \mid s, \pi_k(s)) V^{\pi_k}(s')$$
- **Policy Improvement:**
  For all $s \in \mathcal{S}$:
  $$\pi_{k+1}(s) = \arg\max_{a \in \mathcal{A}} \left[R(s, a) + \gamma \sum_{s'} P(s' \mid s, a) V^{\pi_k}(s') \right]$$

until $\pi_{k+1} = \pi_k$

return π_k

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

Policy Evaluation with Markov Chains

Fixing a Policy Induces a Markov Chain: Given a policy π_k, the MDP reduces to a Markov chain with rewards:

$$\mathcal{P}^{\pi_k}(s' \mid s) = \mathcal{P}(s' \mid s, \pi_k(s)), \quad \mathcal{R}^{\pi_k}(s) = \mathcal{R}(s, \pi_k(s))$$

Policy Evaluation = Computing Value of the Chain

$$V^{\pi_k}(s) = R^{\pi_k}(s) + \gamma \sum_{s'} P^{\pi_k}(s' \mid s)\, V^{\pi_k}(s')$$

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

Value Iteration

In policy iteration, a **policy is explicitly maintained**:

- evaluate π_k
- then improve π_k

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

Value Iteration

In policy iteration, a **policy is explicitly maintained**:

- evaluate π_k
- then improve π_k

In value iteration, we **do not maintain a policy explicitly**.

Instead, we directly update the **value function** using the Bellman optimality equation:

$$V_{k+1}(s) = \max_a \sum_{s'} \mathcal{P}(s'|s,a) \left[\mathcal{R}(s,a,s') + \gamma V_k(s')\right]$$

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

Value Iteration

From IAI:

function VALUE-ITERATION(mdp, ϵ) **returns** a utility function
  inputs: mdp, an MDP with states S, actions $A(s)$, transition model $P(s' \mid s, a)$,
    rewards $R(s, a, s')$, discount γ
    ϵ, the maximum error allowed in the utility of any state
  local variables: U, U', vectors of utilities for states in S, initially zero
    δ, the maximum relative change in the utility of any state

  repeat
    $U \leftarrow U'$; $\delta \leftarrow 0$
    for each state s **in** S **do**
      $U'[s] \leftarrow \max_{a \in A(s)}$ Q-VALUE(mdp, s, a, U)
      if $|U'[s] - U[s]| > \delta$ **then** $\delta \leftarrow |U'[s] - U[s]|$
  until $\delta \leq \epsilon(1 - \gamma)/\gamma$
  return U

What is Stochastic Planning?

Challenges

Solution Approaches

The Bellman Equation

Dynamic Programming Methods

Heuristic Search in Stochastic Domains

Conclusion

Value Iteration

Value Iteration (input: MDP)

Initialize $V_0(s)$ arbitrarily for all $s \in \mathcal{S}$

repeat

- **Value Update:**
  For all $s \in \mathcal{S}$:

$$V_{k+1}(s) = \max_{a \in \mathcal{A}} \left[R(s,a) + \gamma \sum_{s'} P(s' \mid s,a) V_k(s') \right]$$

until $\max_s |V_{k+1}(s) - V_k(s)| < \epsilon$

Extract policy:

$$\pi^*(s) = \arg\max_{a \in \mathcal{A}} \left[R(s,a) + \gamma \sum_{s'} P(s' \mid s,a) V^*(s') \right]$$

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

Value Iteration

Key Idea:

- Each update performs:
  - a **one-step lookahead**
  - followed by a **greedy choice**
- The value function is pushed toward optimality:

$$V_k \longrightarrow V^*$$

- A policy can be extracted at any time:

$$\pi_k(s) = \arg\max_a Q_k(s, a)$$

- In practice:
  - the policy often stabilizes **before** values converge.

What does this mean?

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

Value Iteration

Policy stabilization before value Convergence : Optimal decisions may emerge before exact values are known

- Value iteration refines estimates:

$$V_k(s) \longrightarrow V^*(s)$$

- But the policy depends only on the **best action**:

$$\pi_k(s) = \arg\max_a Q_k(s, a)$$

- Once the **ordering of actions** is correct:
  - the greedy policy stops changing
  - even if values are still slightly inaccurate
- **Early stopping:**
  - stop when the policy stabilizes: $\pi_k = \pi_{k+1}$
  - avoids unnecessary value updates

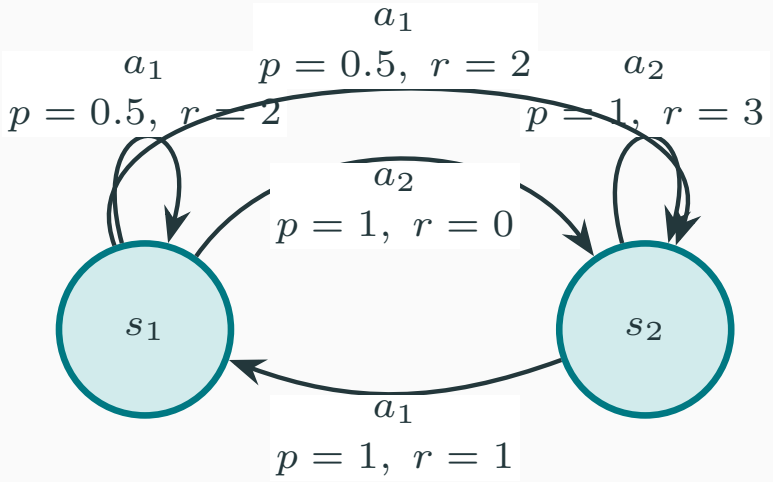
What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

MDP Example

States: $\mathcal{S} = \{s_1, s_2\}$

Actions: $\mathcal{A} = \{a_1, a_2\}$

Discount: $\gamma = 0.9$



Transitions and Rewards:

s	a	s'	$P(s'\|s,a)$	$R(s,a)$
s_1	a_1	s_1	0.5	2
s_1	a_1	s_2	0.5	2
s_1	a_2	s_2	1.0	0
s_2	a_1	s_1	1.0	1
s_2	a_2	s_2	1.0	3

What is Stochastic Planning?

Challenges

Solution Approaches

The Bellman Equation

Dynamic Programming Methods

Heuristic Search in Stochastic Domains

Conclusion

MDP Example: Policy Iteration

Initial policy:

$$\pi_0(s_1) = a_1, \quad \pi_0(s_2) = a_1$$

Initial values:

$$V_0^{\pi 0}(s_1) = 0, \quad V_0^{\pi 0}(s_2) = 0$$

Bellman expectation equation:

$$V^{\pi}(s) = R(s, \pi(s)) + \gamma \sum_{s'} P(s'|s, \pi(s)) V^{\pi}(s')$$

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

MDP Example: Policy Iteration

Policy Evaluation: **Iteration 1**

At s_1:

$$V_1^{\pi_0}(s_1) = 2 + 0.9 \cdot (0.5 \cdot 0 + 0.5 \cdot 0) = 2$$

At s_2:

$$V_1^{\pi_0}(s_2) = 1 + 0.9 \cdot 0 = 1$$

Values:

$$V_1^{\pi_0} = (2,\ 1)$$

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

MDP Example: Policy Iteration

Policy Evaluation: **Iteration 2**

At s_1:

$$V_2^{\pi 0}(s_1) = 2 + 0.9 \cdot (0.5 \cdot 2 + 0.5 \cdot 1) = 3.35$$

At s_2:

$$V_2^{\pi 0}(s_2) = 1 + 0.9 \cdot 2 = 2.8$$

Values:

$$V_2^{\pi 0} = (3.35,\ 2.8)$$

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

MDP Example: Policy Iteration

Policy Evaluation- Solve:

$$V^{\pi_0}(s_1) = 2 + 0.9(0.5V^{\pi_0}(s_1) + 0.5V^{\pi_0}(s_2))$$

$$V^{\pi_0}(s_2) = 1 + 0.9V^{\pi_0}(s_1)$$

Converged solution:

$$V^{\pi_0}(s_1) \approx 20, \quad V^{\pi_0}(s_2) \approx 19$$

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

MDP Example: Policy Iteration

Policy Improvement: **Greedy improvement:**

$$\pi_1(s) = \arg\max_a Q^{\pi_0}(s, a)$$

At s_1:

$$Q^{\pi_0}(s_1, a_1) = 2 + 0.9(0.5 \cdot 20 + 0.5 \cdot 19) = 19.55$$

$$Q^{\pi_0}(s_1, a_2) = 0 + 0.9 \cdot 19 = 17.1$$

$$\Rightarrow \pi_1(s_1) = a_1$$

At s_2:

$$Q^{\pi_0}(s_2, a_1) = 1 + 0.9 \cdot 20 = 19$$

$$Q^{\pi_0}(s_2, a_2) = 3 + 0.9 \cdot 19 = 20.1$$

$$\Rightarrow \pi_1(s_2) = a_2$$

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

MDP Example: Policy Iteration

Final Step (Convergence)

Current policy: $\pi_1(s_1) = a_1, \quad \pi_1(s_2) = a_2$

Evaluate π_1:

$$V^{\pi_1}(s_1) = 2 + 0.9(0.5V^{\pi_1}(s_1) + 0.5V^{\pi_1}(s_2))$$

$$V^{\pi_1}(s_2) = 3 + 0.9V^{\pi_1}(s_2)$$

Solve:

$$V^{\pi_1}(s_2) = \frac{3}{1 - 0.9} = 30$$

$$V^{\pi_1}(s_1) = 2 + 0.9(0.5V^{\pi_1}(s_1) + 0.5 \cdot 30) = 29$$

Policy improvement:

Both s_1: a_1, s_2: a_2 remain optimal

Policy unchanged $\Rightarrow$ **optimal policy found**

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

MDP Example: Value Iteration

Initialize:

$$V_0(s_1) = 0, \qquad V_0(s_2) = 0$$

At each iteration:

$$V_{k+1}(s) = \max_{a \in \mathcal{A}} \left[R(s,a) + \gamma \sum_{s'} P(s' \mid s,a) V_k(s') \right]$$

For this MDP:

$$V_{k+1}(s_1) = \max \left\{ 2 + 0.9\big(0.5V_k(s_1) + 0.5V_k(s_2)\big),\ 0 + 0.9V_k(s_2) \right\}$$

$$V_{k+1}(s_2) = \max \left\{ 1 + 0.9V_k(s_1),\ 3 + 0.9V_k(s_2) \right\}$$

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

MDP Example: Value Iteration

Iteration 1 - Start with:

$$V_0(s_1) = 0, \qquad V_0(s_2) = 0$$

At s_1:

$$Q_0(s_1, a_1) = 2 + 0.9(0.5 \cdot 0 + 0.5 \cdot 0) = 2$$

$$Q_0(s_1, a_2) = 0 + 0.9 \cdot 0 = 0$$

$$V_1(s_1) = \max\{2, 0\} = 2$$

At s_2:

$$Q_0(s_2, a_1) = 1 + 0.9 \cdot 0 = 1$$

$$Q_0(s_2, a_2) = 3 + 0.9 \cdot 0 = 3$$

$$V_1(s_2) = \max\{1, 3\} = 3$$

$$V_1 = (2, 3)$$

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

MDP Example: Value Iteration

Iteration 2 - using $V_1 = (2, 3)$

At s_1:

$$Q_1(s_1, a_1) = 2 + 0.9(0.5 \cdot 2 + 0.5 \cdot 3) = 2 + 0.9 \cdot 2.5 = 4.25$$

$$Q_1(s_1, a_2) = 0 + 0.9 \cdot 3 = 2.7$$

$$V_2(s_1) = \max\{4.25, 2.7\} = 4.25$$

At s_2:

$$Q_1(s_2, a_1) = 1 + 0.9 \cdot 2 = 2.8$$

$$Q_1(s_2, a_2) = 3 + 0.9 \cdot 3 = 5.7$$

$$V_2(s_2) = \max\{2.8, 5.7\} = 5.7$$

$$V_2 = (4.25, 5.7)$$

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

MDP Example: Value Iteration

Iteration 3 - Using:

$$V_2 = (4.25, 5.7)$$

At s_1:

$$Q_2(s_1, a_1) = 2 + 0.9(0.5 \cdot 4.25 + 0.5 \cdot 5.7)$$

$$= 2 + 0.9 \cdot 4.975 = 6.4775$$

$$Q_2(s_1, a_2) = 0 + 0.9 \cdot 5.7 = 5.13$$

$$V_3(s_1) = \max\{6.4775, 5.13\} = 6.4775$$

At s_2:

$$Q_2(s_2, a_1) = 1 + 0.9 \cdot 4.25 = 4.825$$

$$Q_2(s_2, a_2) = 3 + 0.9 \cdot 5.7 = 8.13$$

$$V_3(s_2) = \max\{4.825, 8.13\} = 8.13$$

$$V_3 = (6.4775, 8.13)$$

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

MDP Example: Value Iteration

Value Evolution:

k	$V_k(s_1)$	$V_k(s_2)$
0	0	0
1	2	3
2	4.25	5.7
3	6.48	8.13
4	8.57	10.32
5	10.51	12.29
⋮	⋮	⋮

The values gradually approach the optimal values.

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

MDP Example: Value Iteration

At each iteration, value iteration induces a greedy policy:

$$\pi_k(s) = \arg\max_a Q_k(s, a)$$

From the updates above:

$$\pi_1(s_1) = a_1, \qquad \pi_1(s_2) = a_2$$

$$\pi_2(s_1) = a_1, \qquad \pi_2(s_2) = a_2$$

$$\pi_3(s_1) = a_1, \qquad \pi_3(s_2) = a_2$$

At convergence:

$$V^*(s_1) = 2 + 0.9\big(0.5V^*(s_1) + 0.5V^*(s_2)\big)$$

$$V^*(s_2) = 3 + 0.9V^*(s_2)$$

$$\boxed{V^*(s_1) \approx 28.18, \qquad V^*(s_2) = 30}$$

The optimal actions are: $\pi^*(s_1) = a_1, \qquad \pi^*(s_2) = a_2$

In this example, the greedy policy stabilizes early, even while the values keep changing

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

Complexity: Policy Iteration vs. Value Iteration

Value iteration and policy iteration converge to the same optimal values.

Component	Policy Iteration	Value Iteration
Policy Evaluation	Solve n equations Direct: $\mathcal{O}(n^3)$ Iterative: $\mathcal{O}(Kn^2\|\mathcal{A}\|)$	—
Update Step	$\mathcal{O}(n^2\|\mathcal{A}\|)$	Per state: $\mathcal{O}(n\|\mathcal{A}\|)$ Total: $\mathcal{O}(n^2\|\mathcal{A}\|)$
Iterations	Typically small	Often larger depending on γ, ϵ
Total Complexity	$\mathcal{O}(K_{\text{PI}} \cdot n^2\|\mathcal{A}\|)$	$\mathcal{O}(K \cdot n^2\|\mathcal{A}\|)$

Policy Iteration: few expensive iterations
Value Iteration: many cheap iterations

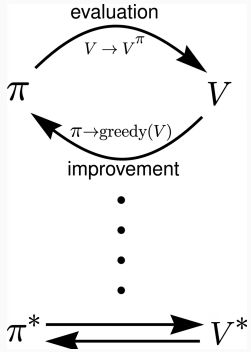
What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

Complexity: Policy Iteration vs. Value Iteration

- **Value Iteration**
  - Stops when: $\max_s |V_{k+1}(s) - V_k(s)| < \epsilon$
  - Contraction with factor γ
  - Number of iterations:
$$K = O\left(\frac{\log(1/\epsilon)}{1-\gamma}\right)$$
- **Policy Iteration**
  - Each iteration strictly improves policy
  - Worst case: $\leq |\mathcal{A}|^{|\mathcal{S}|}$
  - **In practice:**
    - very few iterations (often < 10)
- **Modified / Iterative Policy Iteration**: Perform **partial policy evaluation**.
  - Trade-off between PI and VI: fewer iterations than VI, cheaper iterations than PI.
  - Often best in practice.

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

Unifying Perspective: Generalized Policy Iteration



- Policy Iteration:
  - evaluation until convergence
  - then full improvement
- Value Iteration:
  - one-step evaluation
  - immediate improvement

Evaluation $\longleftrightarrow$ Improvement

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

Heuristic Search in Stochastic Domains

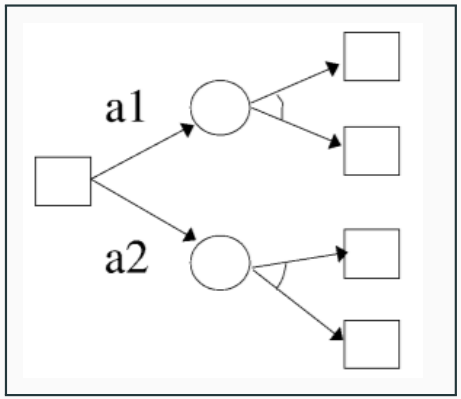
Heuristic Search in AND/OR Graphs

- Extend heuristic search to **stochastic / nondeterministic domains** where problems are represented as **AND–OR graph**.
- **Heuristic function** $h(s)$:
  - estimates [expected] cost-to-go from a state
  - guides which branches to expand
- Cost propagation:
  - OR: $\min$ over actions
  - AND: sum (or expectation) over outcomes
- **AO* search:**
  - expands most promising partial solution graph
  - updates values bottom-up via **backpropagation**
  - returns a **conditional plan / policy**

Heuristics + AND–OR structure enable efficient planning under uncertainty

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

Heuristic Search in AND/OR Graphs



A* assumes deterministic actions and finds a sequence of actions (path) leading from the start state to a goal state.

AO* generalizes A* to AND–OR graphs and finds a solution that has a conditional structure.

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

AO*

- AO* (Nilsson 1980) is a heuristic search algorithm that can find optimal solutions for acyclic MDPs (e.g., finite-horizon MDPS) without evaluating the entire state space.
- Thus, it provides a useful alternative to dynamic programming algorithms for MDPs such as value iteration and policy iteration.
- Iterates over two steps:
  - **Expanding** the current best partial policy
  - **Updating** the states' value according to the current policy envelope to propagate values from the newly expanded state.

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

AO*: idea

Maintain:

- cost estimate
- best partial solution graph

Iteratively:

- expand most promising node
- update costs **bottom-up**
- mark solved subgraphs

Terminate when the root is solved

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

AO*: Basic structure

Cost computation:

OR node (state): Max reward:

$$f(n) = \min_{a \in A(n)} [\mathcal{R}(n, a) + f(\text{child via } a)]$$

Min cost:

$$f(n) = \min_{a \in A(n)} [\mathcal{C}(n, a) + f(\text{child via } a)]$$

AND node (action outcomes):

$$f(n) = \sum_i f(n_i)$$

- OR: choose best action
- AND: must account for all outcomes

AO* propagates costs upward until consistency

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

AO*: Example



Initial state

What is Stochastic Planning?

Challenges

Solution Approaches

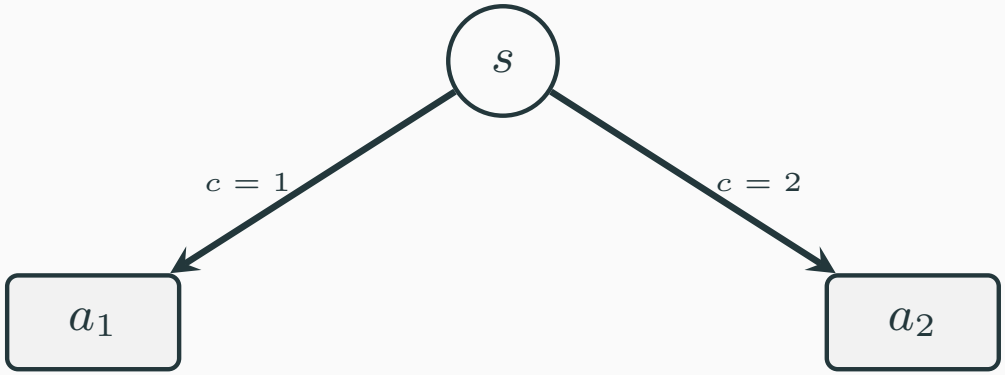
The Bellman Equation

Dynamic Programming Methods

Heuristic Search in Stochastic Domains

Conclusion

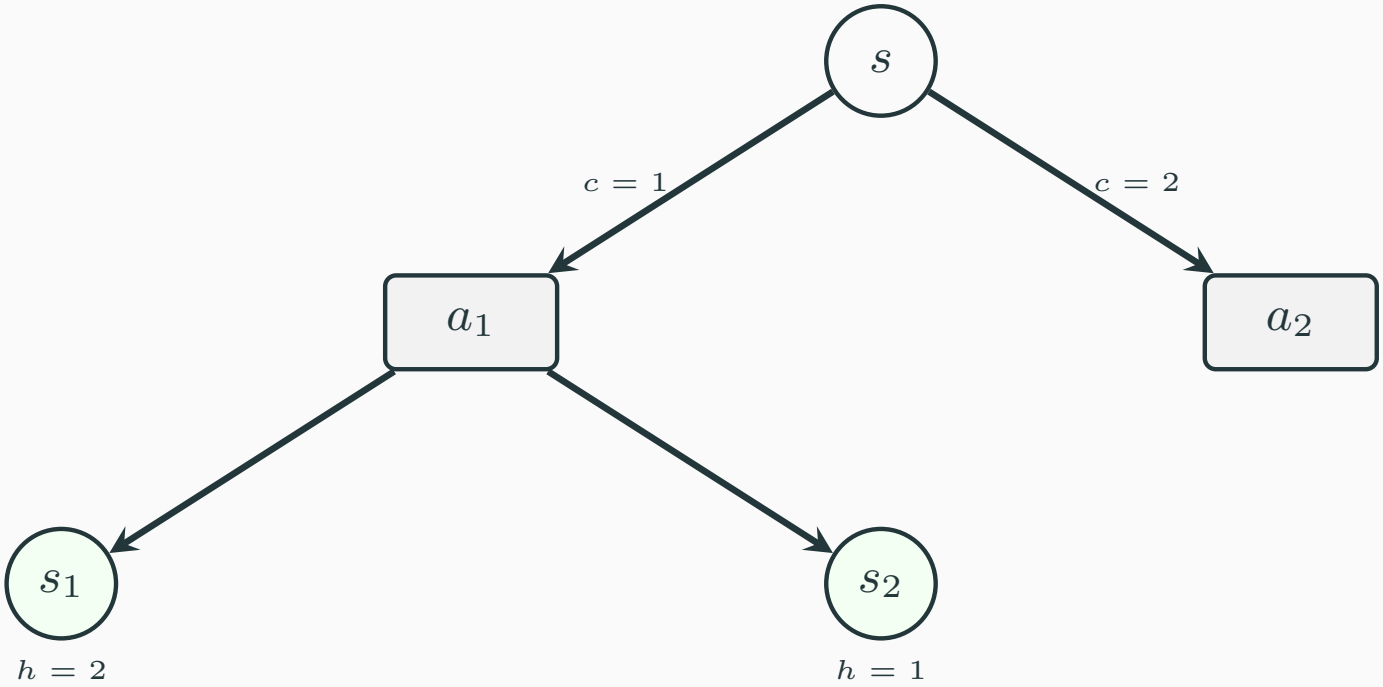
AO*: Example



Expand the root: AO* considers alternative actions

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

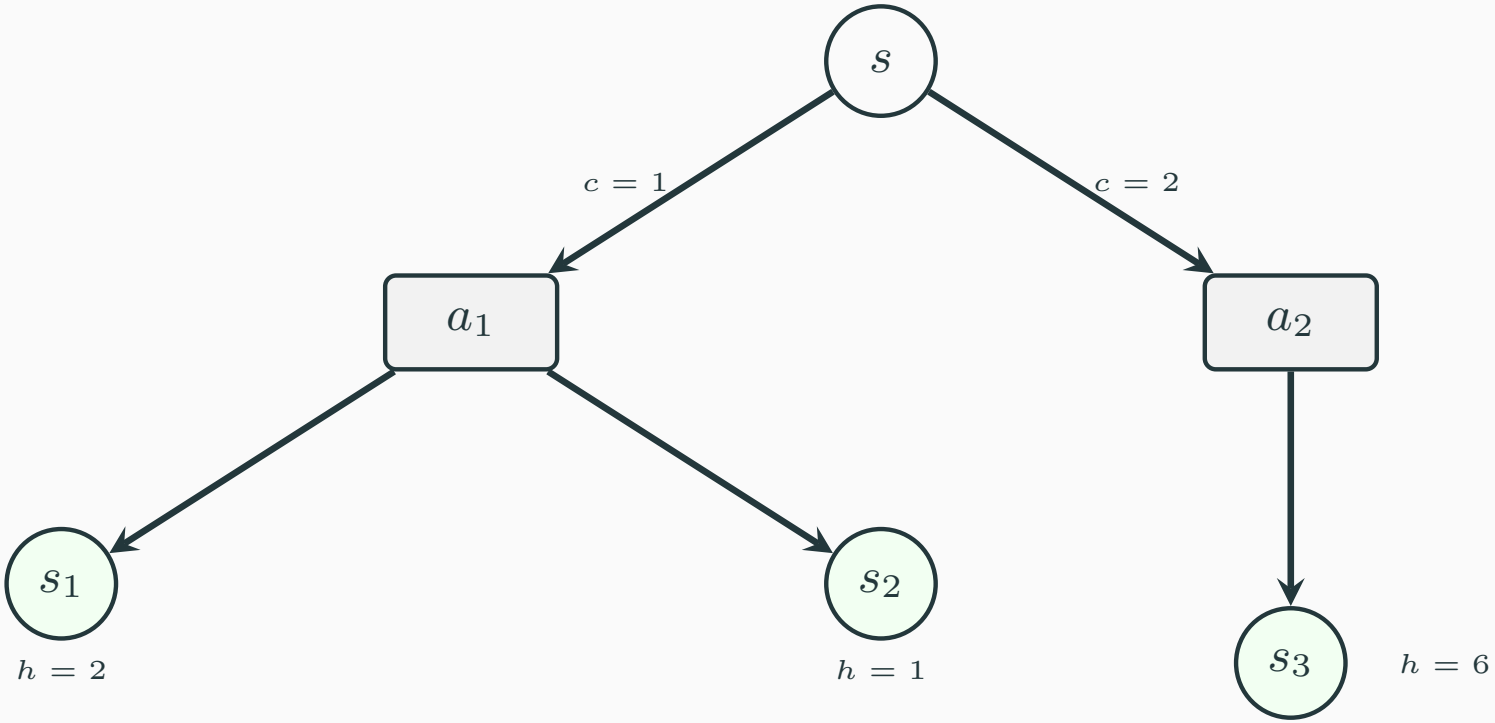
AO*: Example



Expanding a_1: both successor states must be considered (AND)

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

AO*: Example



Evaluate alternatives using heuristics:

$$a_1 : 1 + \big(0.5 \cdot 2 + 0.5 \cdot 1\big) = 1 + 1.5 = 2.5 \quad \text{vs.} \quad a_2 : 2 + 6 = 8$$

What is Stochastic Planning?

Challenges

Solution Approaches

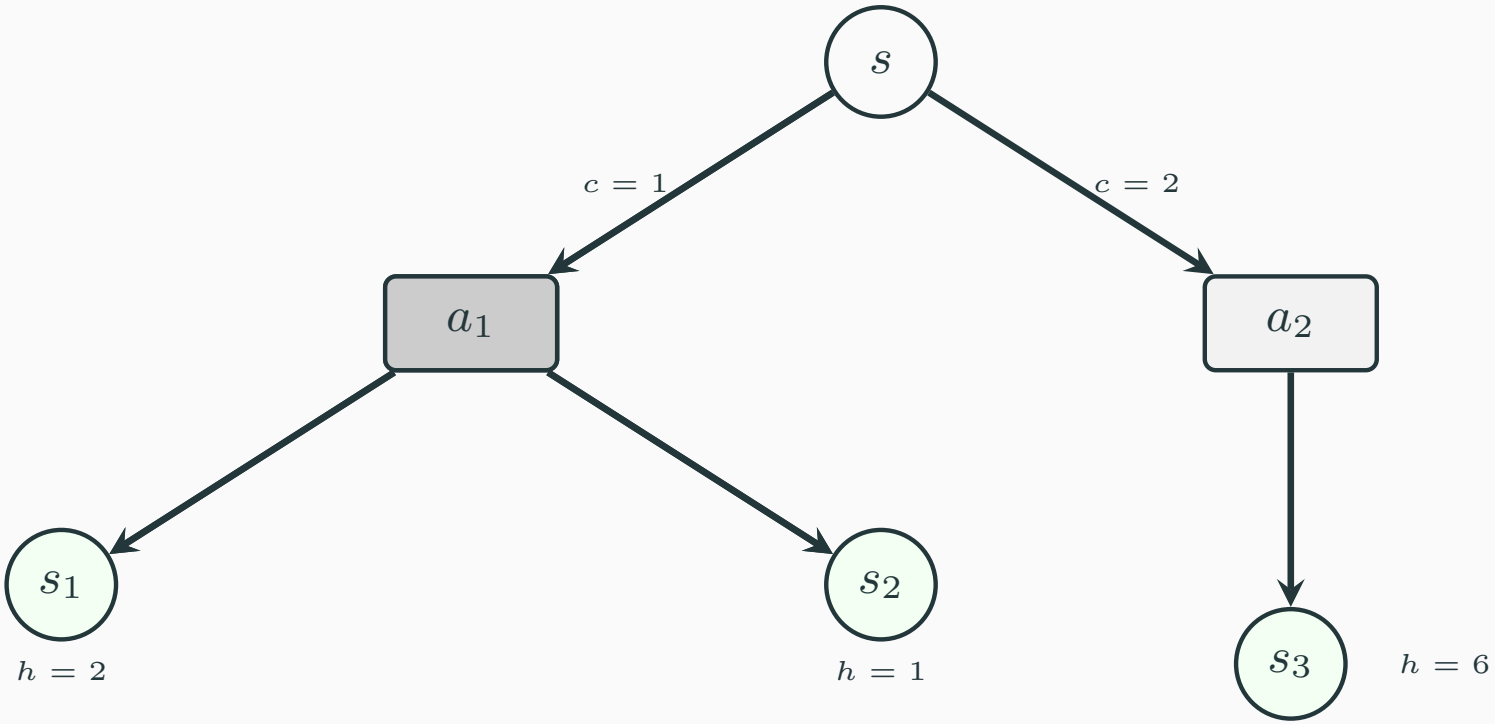
The Bellman Equation

Dynamic Programming Methods

Heuristic Search in Stochastic Domains

Conclusion

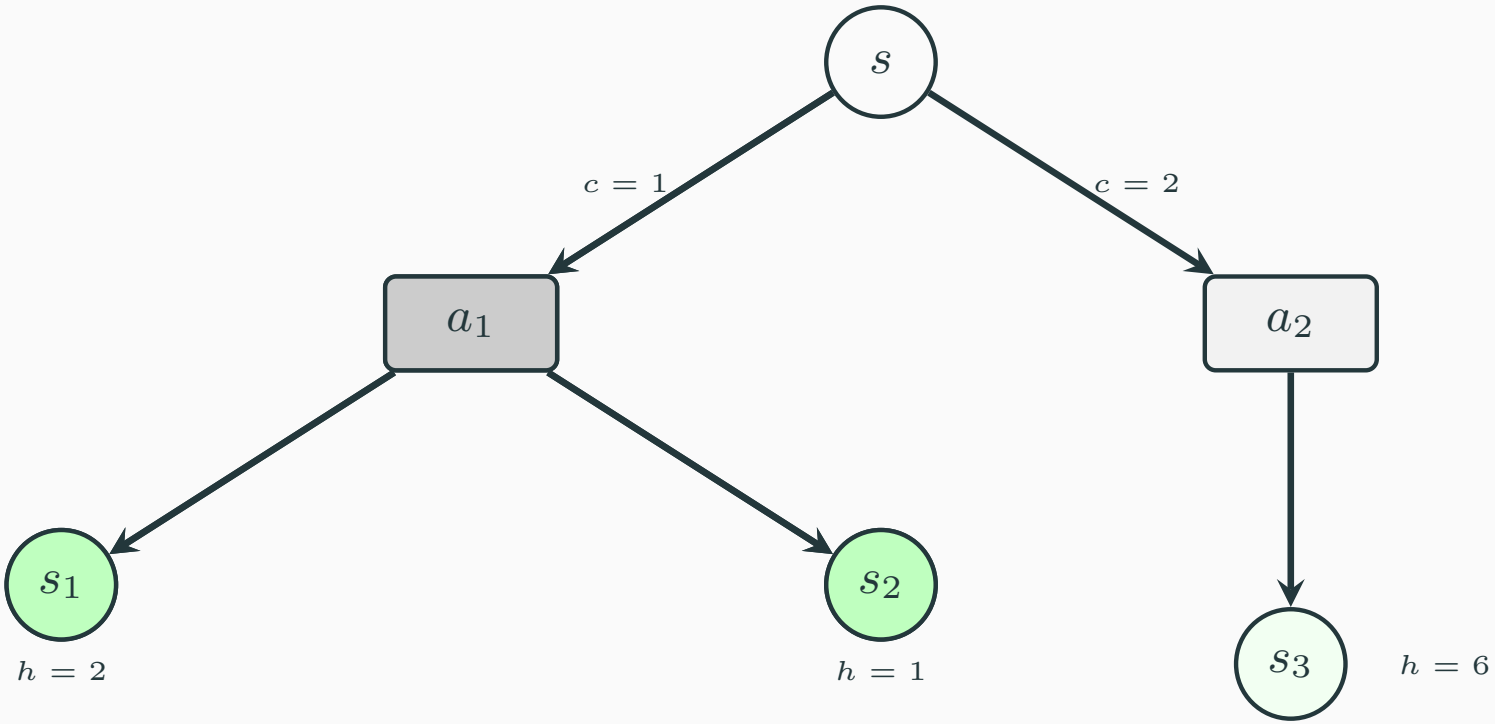
AO*: Example



AO* selects the current best partial solution tree (left branch)

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

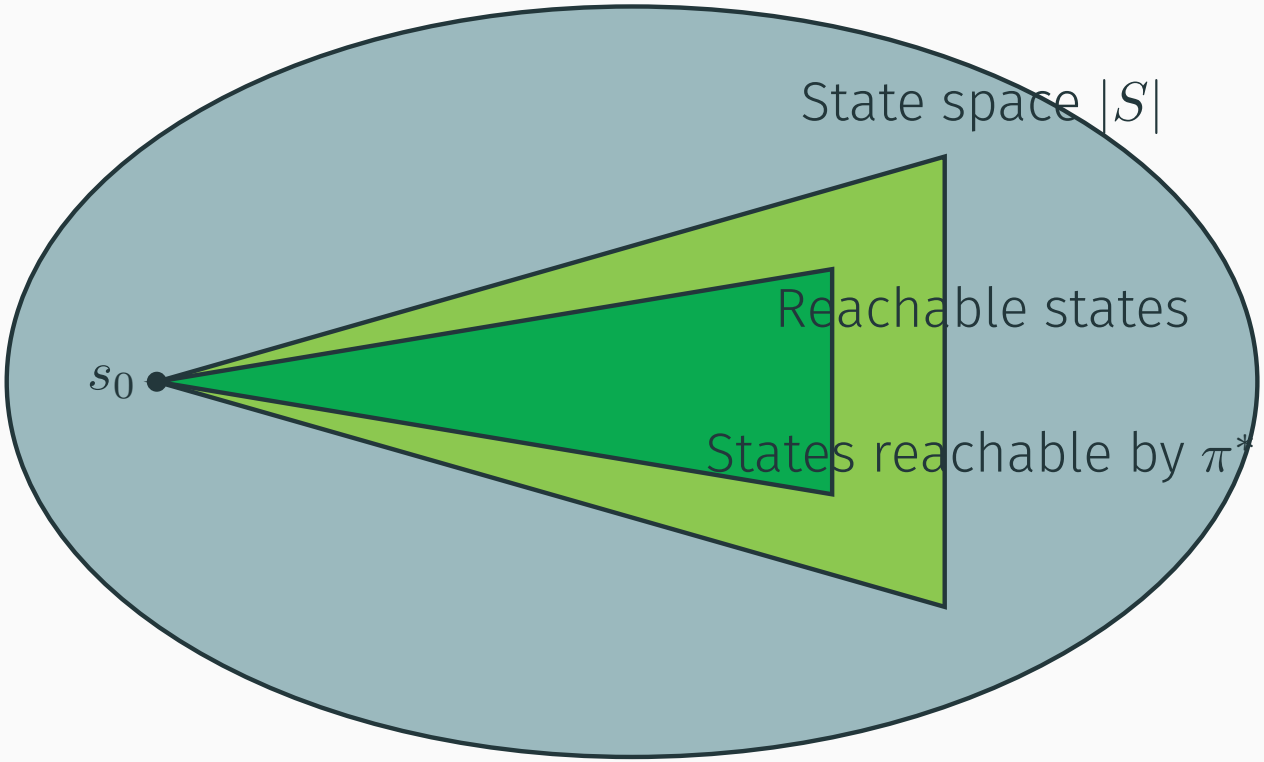
AO*: Example



Once all nodes on the selected branch are expanded, values are propagated to the initial state.

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

AO*



What is the difference between the set of states that must be considered by the solution found by A* vs. AO*?

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

AO* Algorithm (Sketch)

1. Initialize root with heuristic $h(n)$
2. Mark all nodes as UNSOLVED
3. Repeat:
   - follow best partial solution graph
   - expand a frontier node
   - assign heuristic values to children
   - update costs upward
   - mark nodes as solved when all children solved
4. Stop when root is solved

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

AO*

σ_0 := make-root-node(init())
mark all nodes UNKNOWN; $f(\sigma) := h(\sigma)$
open := $\{\sigma_0\}$

while not σ_0 is SOLVED:
  σ := select a leaf node on the current best partial solution graph
  expand σ
  for each successor σ' of σ:
    if σ' is a goal:
      $f(\sigma') := 0$; mark σ' SOLVED
    else:
      $f(\sigma') := h(\sigma')$; mark σ' UNKNOWN

  repeat (back-propagation phase):
    select ancestor σ_p whose value may change
    if σ_p is an OR node:
      choose and mark best action $a^* = \arg\min_a \left(c(\sigma_p, a) + F(\text{Succ}(\sigma_p, a))\right)$
      $f(\sigma_p) := c(\sigma_p, a^*) + F(\text{Succ}(\sigma_p, a^*))$
      if all successors under a^* are SOLVED:
        mark σ_p SOLVED
    if σ_p is an AND node:
      $f(\sigma_p) := \sum_{\sigma' \in \text{Succ}(\sigma_p)} (c(\sigma_p, \sigma') + f(\sigma'))$
      if all successors are SOLVED:
        mark σ_p SOLVED
  until no value changes

return marked solution graph rooted at σ_0

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

AO*

- At each node, AO* stores the **current best choice of children**, this is the **best partial solution graph**, Z.

For OR nodes: $Z(\sigma) = \arg\min_{c \in \text{children}(\sigma)} f(c)$ indicates which single branch is optimal.

For AND nodes: $Z(\sigma) = \text{children}(\sigma)$, specifies that all subproblems must be solved.

Why is Z needed?

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

AO*

- At each node, AO* stores the **current best choice of children**, this is the **best partial solution graph**, Z.

For OR nodes: $Z(\sigma) = \arg\min_{c \in \text{children}(\sigma)} f(c)$ indicates which single branch is optimal.

For AND nodes: $Z(\sigma) = \text{children}(\sigma)$, specifies that all subproblems must be solved.

Why is Z needed?

AO* follows Z links to expand nodes along the current best solution. When costs change, $Z(\sigma)$ is updated and the solution graph is re-routed.

Is AO* optimal?

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

AO*: Optimality

AO* is optimal if:

- The heuristic $h(\sigma)$ is admissible
  - never overestimates the true cost-to-go
- The AND–OR graph is acyclic
- All costs are non-negative

Intuition:

- AO* always expands nodes along the current best solution graph Z
- Cost updates propagate upward
- With an admissible heuristic, the selected solution converges to the optimal one

Limitation: AO* assumes an acyclic and finite AND-OR graph. What happens when the state space contains cycles?

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

LAO*

AO*:

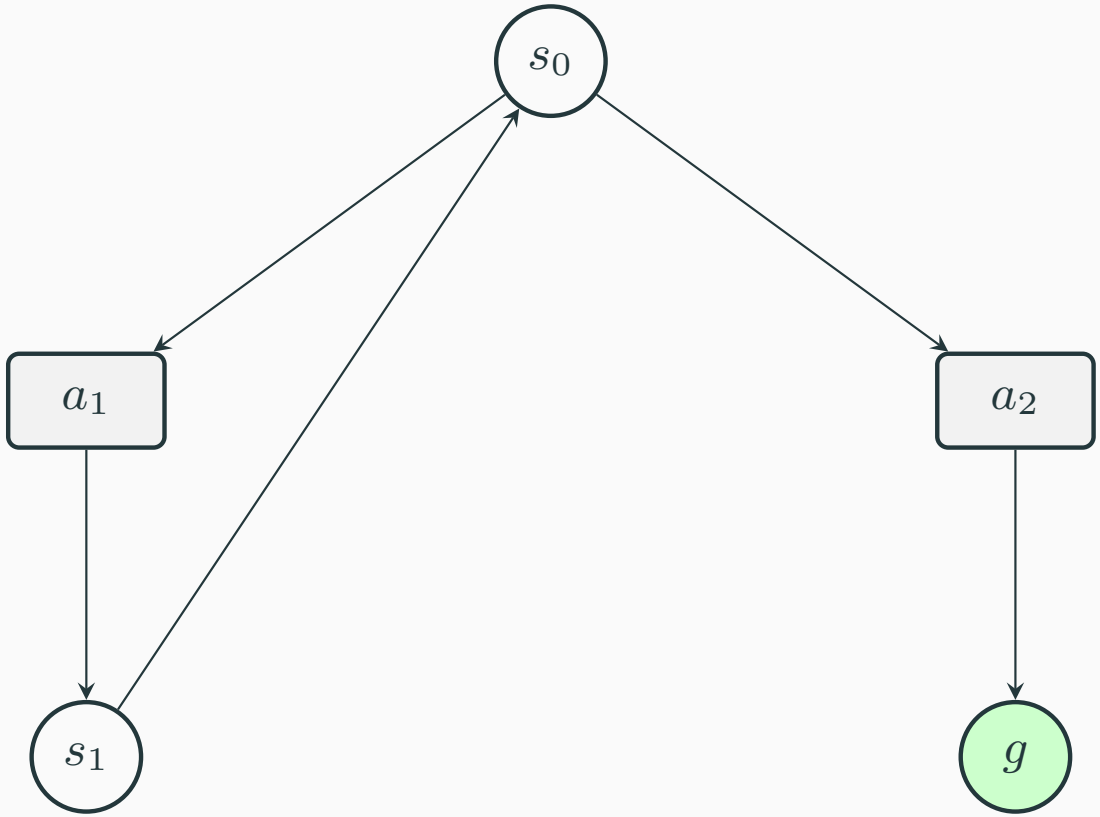
- Designed for **acyclic AND–OR graphs**
- Expands nodes along current best solution graph Z
- Terminates when solution graph reaches goal leaves
- **Cannot handle cycles** (may loop indefinitely)

LAO*:

- Extends AO* to **graphs with cycles**
- Builds a **partial solution graph** reachable from the root
- Uses **dynamic programming (value updates)** over this graph
- Computes a **policy over reachable states**

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

Why AO* Fails on Cycles (and LAO* Succeeds)



- AO* may keep expanding along $s_0 \to s_1 \to s_0 \to \dots$
- The solution graph Z never stabilizes

LAO* idea:

- Treat the graph as a **cyclic MDP** and perform **value updates** over the current partial graph.

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

LAO*

σ_0 := make-root-node(init())
mark all nodes UNEXPANDED; $f(\sigma_0) := h(\sigma_0)$
initialize best partial solution graph Z with root σ_0

while Z contains a nonterminal tip node **or** values have not converged:
  σ := select a nonterminal tip node on the current best partial solution graph Z
  expand σ
  for each successor σ' of σ:
    if σ' is new:
      if σ' is a goal:
        $f(\sigma') := 0$
      else:
        $f(\sigma') := h(\sigma')$
      mark σ' UNEXPANDED

  repeat (dynamic programming phase):
    for each node σ_p in the ancestors of σ and along Z:
      if σ_p is an OR node:
        choose and mark best action
        $a^* = \arg\min_a \left(c(\sigma_p, a) + \sum_{\sigma' \in \text{Succ}(\sigma_p, a)} P(\sigma' \mid \sigma_p, a)\, f(\sigma')\right)$
        $Z(\sigma_p) := \text{Succ}(\sigma_p, a^*)$
        $f(\sigma_p) := c(\sigma_p, a^*) + \sum_{\sigma' \in \text{Succ}(\sigma_p, a^*)} P(\sigma' \mid \sigma_p, a^*)\, f(\sigma')$
    until no value changes by more than ϵ

  update the best partial solution graph Z starting from σ_0

return solution graph / policy induced by Z

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

Heuristics for Stochastic Planning?

Heuristics for Stochastic Planning

How can we guide search when actions have multiple possible outcomes?

- Deterministic planning uses heuristics (e.g., delete relaxation)
- In stochastic domains:
  - actions lead to **multiple possible outcomes**
  - planning becomes an **AND-OR search problem**
- We can use the heuristics we have seen for deterministic domains, but also consider some heuristics that are specific to the non-deterministic case.

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

Admissible Heuristics in Stochastic Domains

Definition (Admissibility)

In **stochastic cost-minimization settings**, a heuristic $h(\sigma)$ is **admissible** if it never overestimates the optimal expected cost-to-go $h(\sigma) \leq V^*(\sigma) \qquad \forall \sigma$

- $V^*(s)$ is the **expected cost** under an optimal policy
- The heuristic must account for **all possible outcomes** of actions

Bellman perspective:

$$V^*(s) = \min_a \left(c(s,a) + \sum_{s'} P(s' \mid s,a)\, V^*(s') \right)$$

An admissible heuristic satisfies:

$$h(s) \leq \min_a \left(c(s,a) + \sum_{s'} P(s' \mid s,a)\, h(s') \right)$$

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

All-Outcome Determinization

- Convert a stochastic problem into a deterministic one:
  - For each stochastic action (s, a), create **one deterministic action per outcome**
  - Reduces AND-OR graph to OR-only search
- Each outcome becomes a separate transition:

$$P(s'|s, a) > 0 \Rightarrow a_{s'} : s \to s'$$

- Solve using standard deterministic planning heuristics
- **Key idea:**
  - ignore probabilities
  - reason over **all possible outcomes**

Is it admissible?

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

All-Outcome Determinization

- Convert a stochastic problem into a deterministic one:
  - For each stochastic action (s, a), create **one deterministic action per outcome**
  - Reduces AND-OR graph to OR-only search
- Each outcome becomes a separate transition:

$$P(s'|s,a) > 0 \;\Rightarrow\; a_{s'} : s \to s'$$

- Solve using standard deterministic planning heuristics
- **Key idea:**
  - ignore probabilities
  - reason over **all possible outcomes**

Is it admissible?

It is admissible, but can lead to overly optimistic heuristics.

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

Most-Likely-Outcome Determinization

- Convert stochastic actions into deterministic ones:
  - For each (s, a)
  - keep only the **most probable outcome**
- Deterministic transition:

$$s \xrightarrow{a} \arg\max_{s'} P(s'|s, a)$$

- Solve using standard deterministic planning heuristics

Is this approach admissible?

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

Conclusion

Evaluation

- **Classical considerations:** completeness, optimality, time complexity, space complexity, anytime guarantees
- **Modern considerations:** robustness, scalability, adaptivity, generality, interpretability, energy efficiency

Where do the methods we have seen stand?

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

Evaluation

- **Classical considerations:** completeness, optimality, time complexity, space complexity, anytime guarantees
- **Modern considerations:** robustness, scalability, adaptivity, generality, interpretability, energy efficiency

Where do the methods we have seen stand?

- The methods we have seen (VI, PI, AO*, LAO*) are **classical model-based algorithms**
- They typically provide strong guarantees:
  - **complete** and **optimal** (under standard assumptions)
- Since they rely on an explicit model:
  - inherently **interpretable**
- However:
  - limited **robustness** to model errors
  - limited **adaptivity** to changes in the environment

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion

Why all this matters nevertheless (especially for RL) ?

Planning algorithms solve decision problems when the **model is known**

- transitions and rewards are given

Reinforcement Learning addresses an extended setting:

- model is not fully unknown
- agent must **learn** from interaction

Core connection:

- Bellman equations $\rightarrow$ foundation of RL
- The methods we have explored are the basis for RL methods

What is Stochastic Planning?
Challenges
Solution Approaches
The Bellman Equation
Dynamic Programming Methods
Heuristic Search in Stochastic Domains
Conclusion